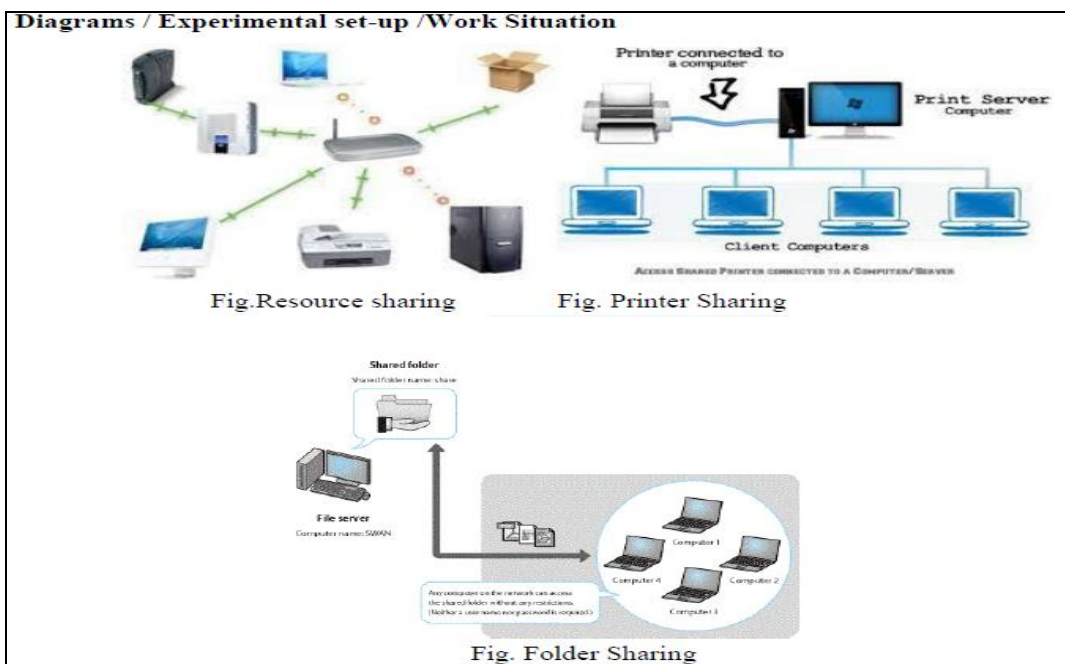


Expt.No: 15. Build a physical Ethernet LAN Network and demonstrate file sharing, printer sharing.

- A resource, or system resource, is any physical or virtual component of limited availability within a computer system.
- Every device connected to a computer system is a resource and every internal system component is also a resource.
- Major resource types are CPU time, Random Access Memory, Hard disk space, Network throughput, Electrical power, External Devices, Input/output operations.
- Virtual system resources include files, network connections and memory areas, whereas a physical resource includes printer, scanner, fax machine etc.
- Types of System Resources 1. Physical 2. Virtual
- Types of Physical Resources
1. Printer
2. Scanner
3. Fax Machine
- Types of Virtual Resources
1. Memory
2. Files
3. CPU time

Resource Sharing

- A shared resource or network share is a device or piece of information on a computer that can be remotely accessed from another computer typically via a local area network or an enterprise Intranet, transparently as if it were a resource in the local machine.
- Examples are shared file access (also known as disk sharing and folder sharing), shared printer access (printer sharing), shared scanner access, etc.
- Resource sharing means reduction in hardware costs. Shared files mean reduction in memory requirement, which indirectly means reduction in file storage expenses.
- **Printer sharing is a feature which allows you to access and use a printer from other computers in network.**

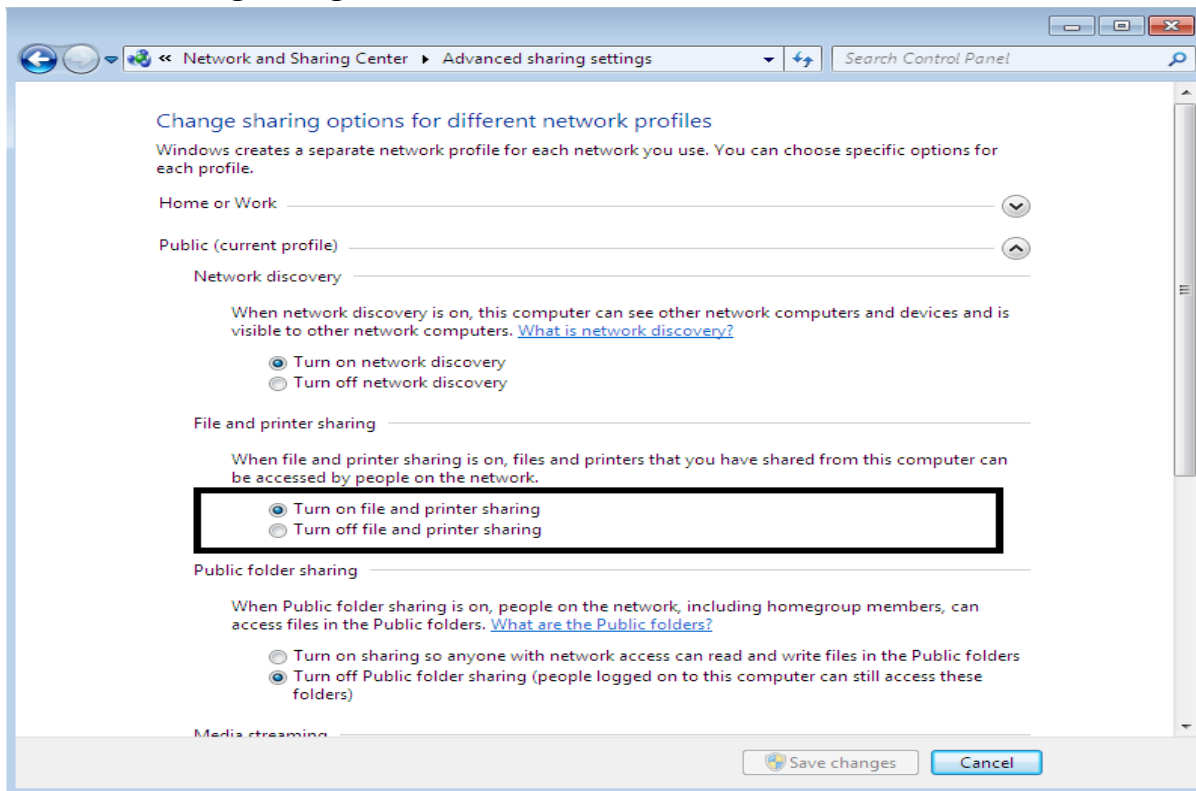


Procedure:

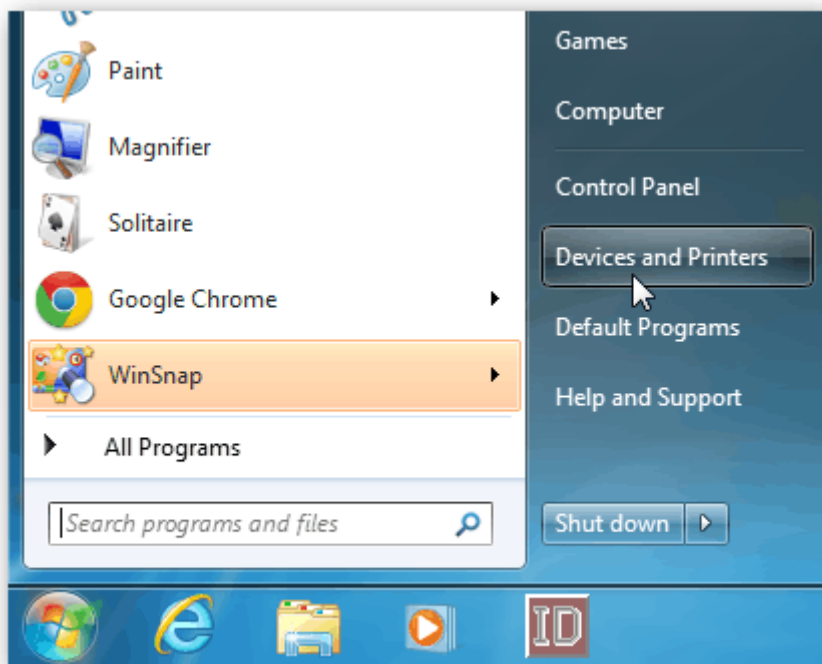
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Share Printer and folder

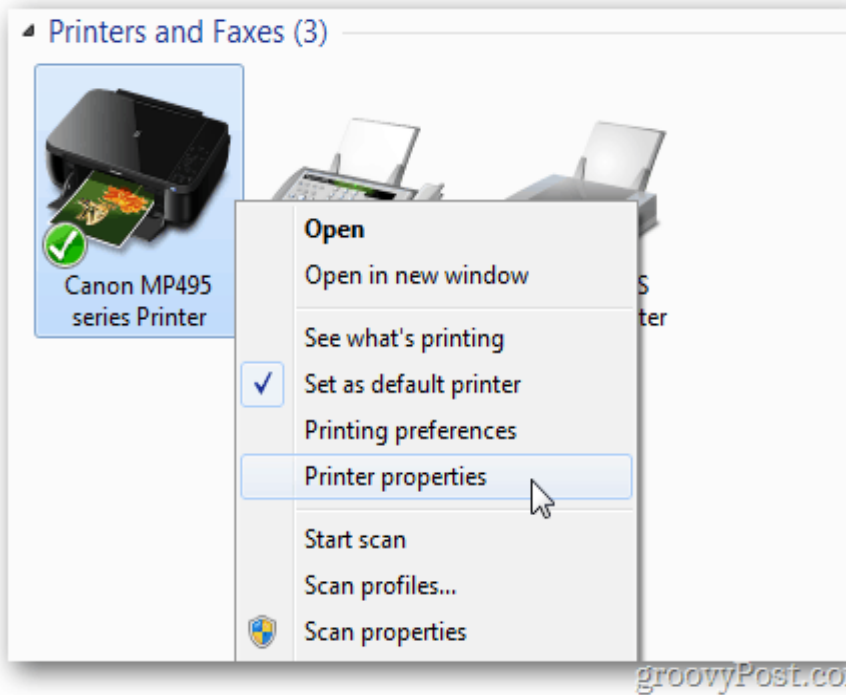
Click on Start Button->Click on Control Panel-> Click on Network and Sharing Center->click on **Change advanced sharing settings**



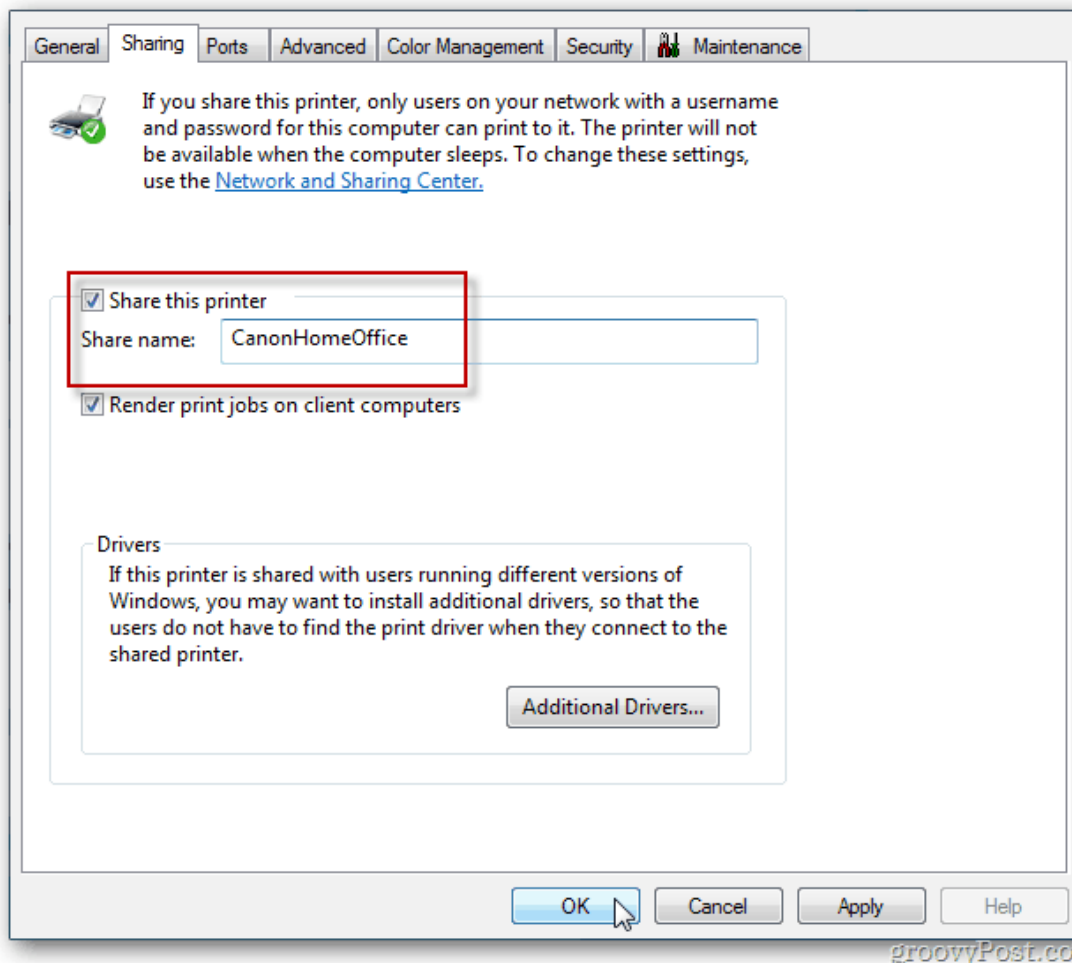
First start with the computer the printer is connected to. Make sure it's installed correctly with the latest drivers. click Start >> Devices and Printers.



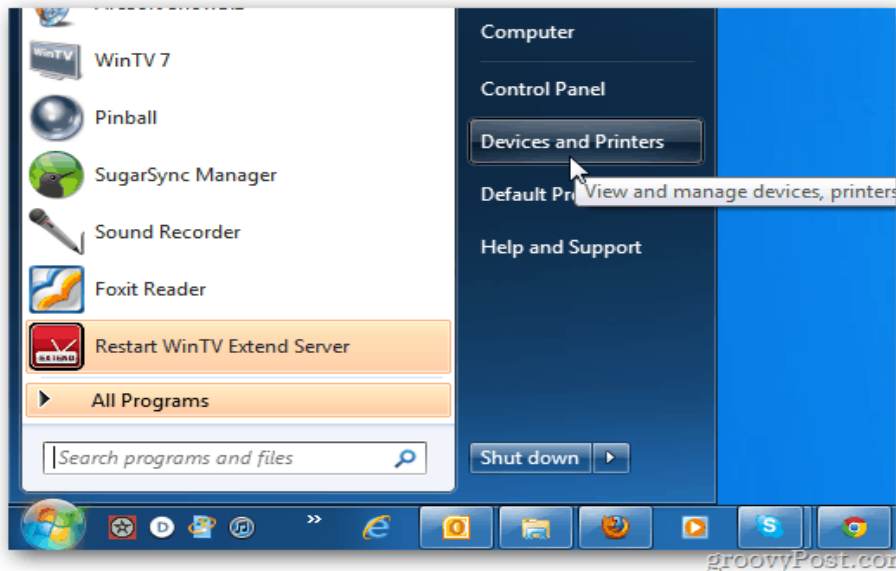
Next, right-click on the printer you want to share and select Printer Properties.



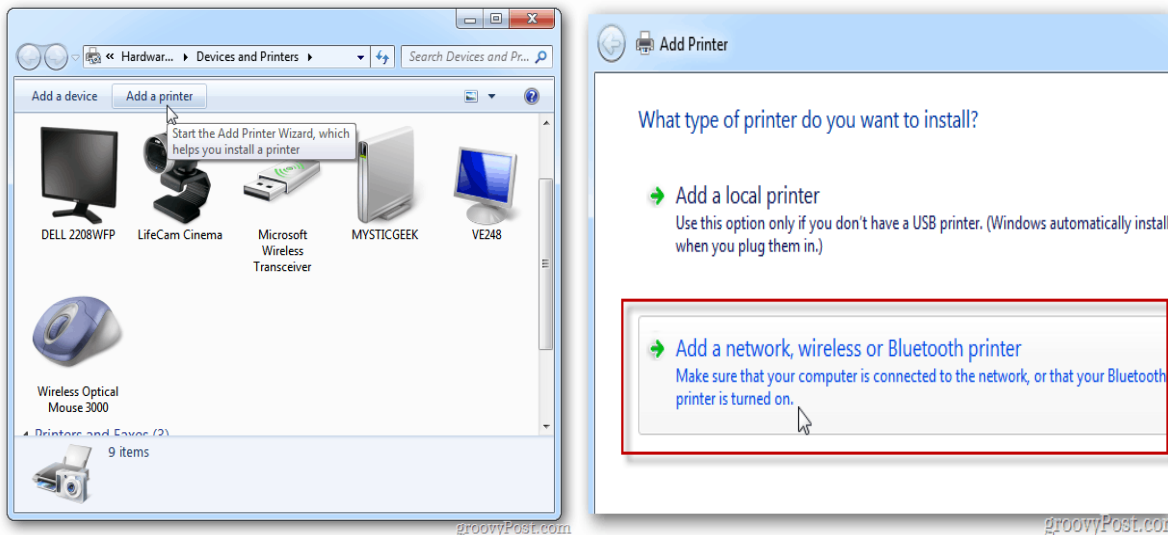
Click the Sharing tab. Make sure **Share this Printer is checked** and give it an easy to remember share name. Click OK.



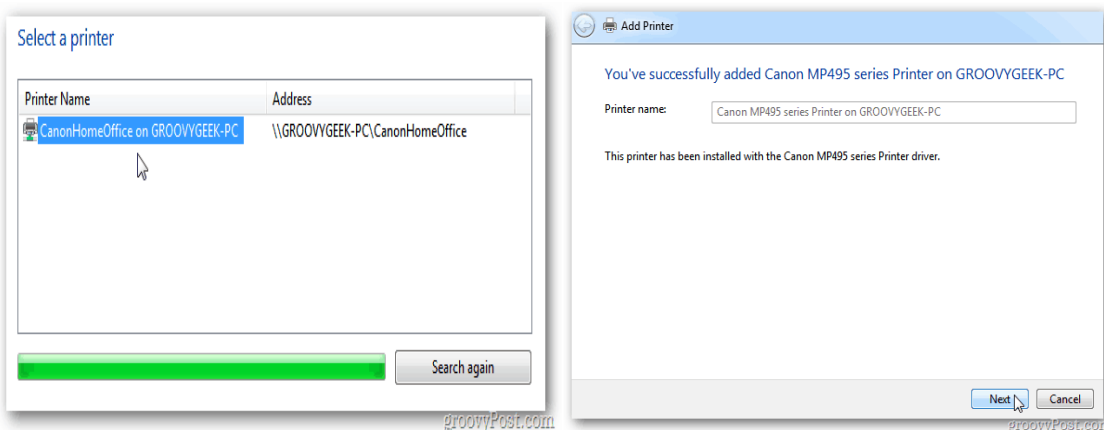
The computer printer is attached need to be powered on to find and print to it.
Now go to the other computer you want to print from. Click *Start >> Devices and Printers*.



Click Add a Printer. Next, click Add a Network, Wireless or Bluetooth Printer.



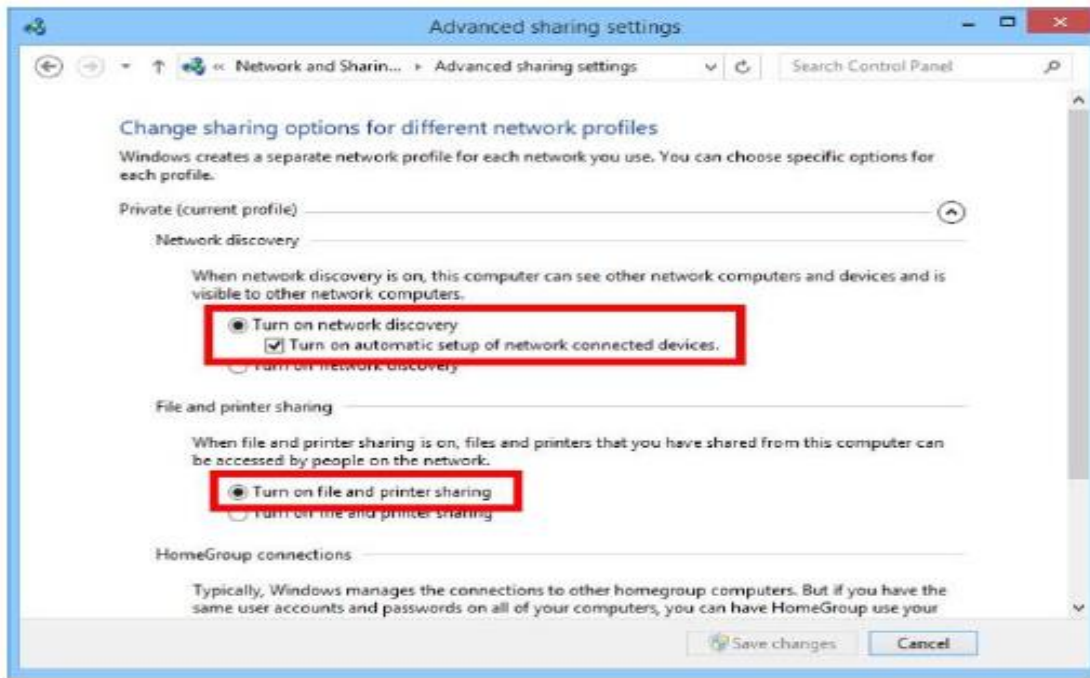
The system will search your network for the shared printer. When it finds the printer, highlight it and click Next. Success. Click Next.



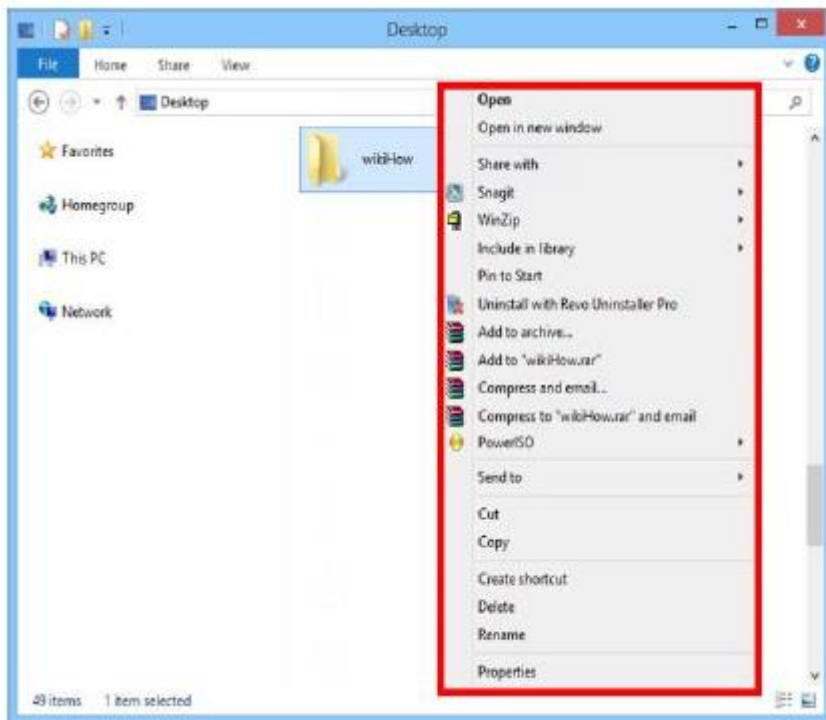
Sharing Specific Folders

Windows 7 - Click the Start button, type "control panel", and press $\leftarrow$ Enter. Double-click the "Network and Sharing Center" icon. Click the "Change advanced sharing settings" link. Expand the profile that you want

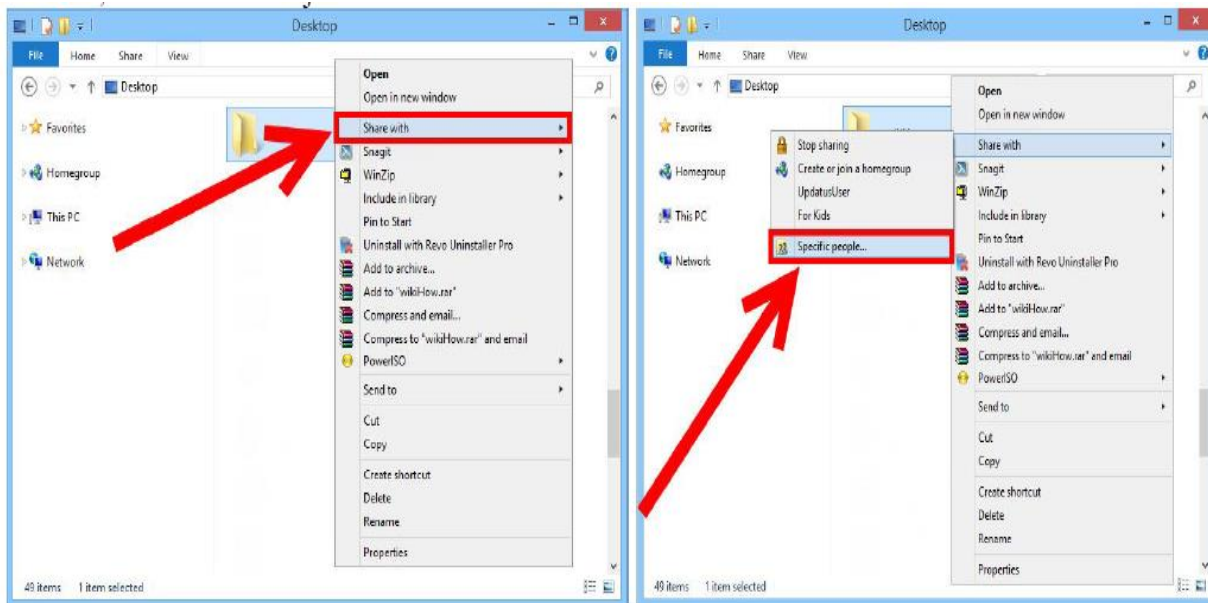
to enable sharing on (Home/Work or Public). Turn on both "**Network discovery**" and "**File and printer sharing**". Click the "Save changes" button and enter your administrator password if necessary



Find the folder you wish to share. Once File and Printer Sharing has been enabled, you can share any folder on your hard drive with other people on your network. Navigate to the folder that you want to share using Explorer. Right-click on it.

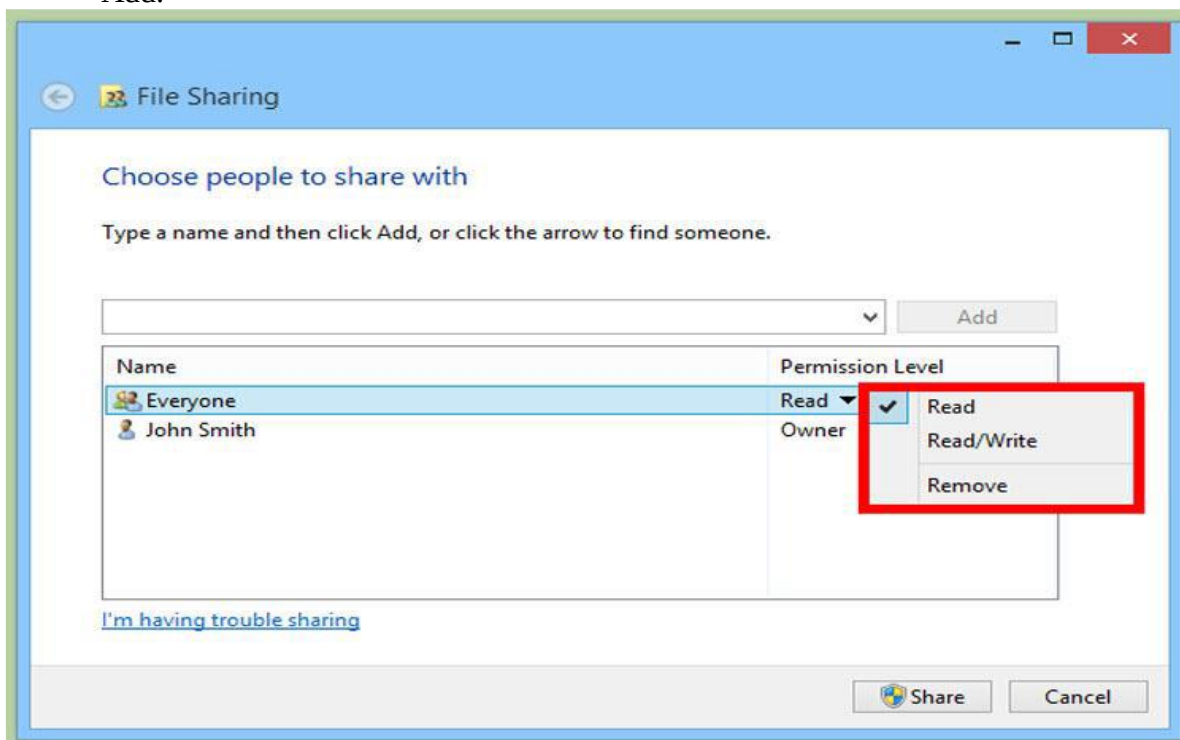


Select the "Share with" option. This will open the Sharing submenu. You can choose to share it with everyone in your Homegroup or select **specific people** to share it with. When choosing a Homegroup option, you can allow other Homegroup members to both read and write to the folder, or limit them to just read from it.



Click the "Specific people" option to select which users you want to share with. This will open a new window with a list of all the users that currently have access to the folder. You can add users to this list and give them specific permissions for the folder.

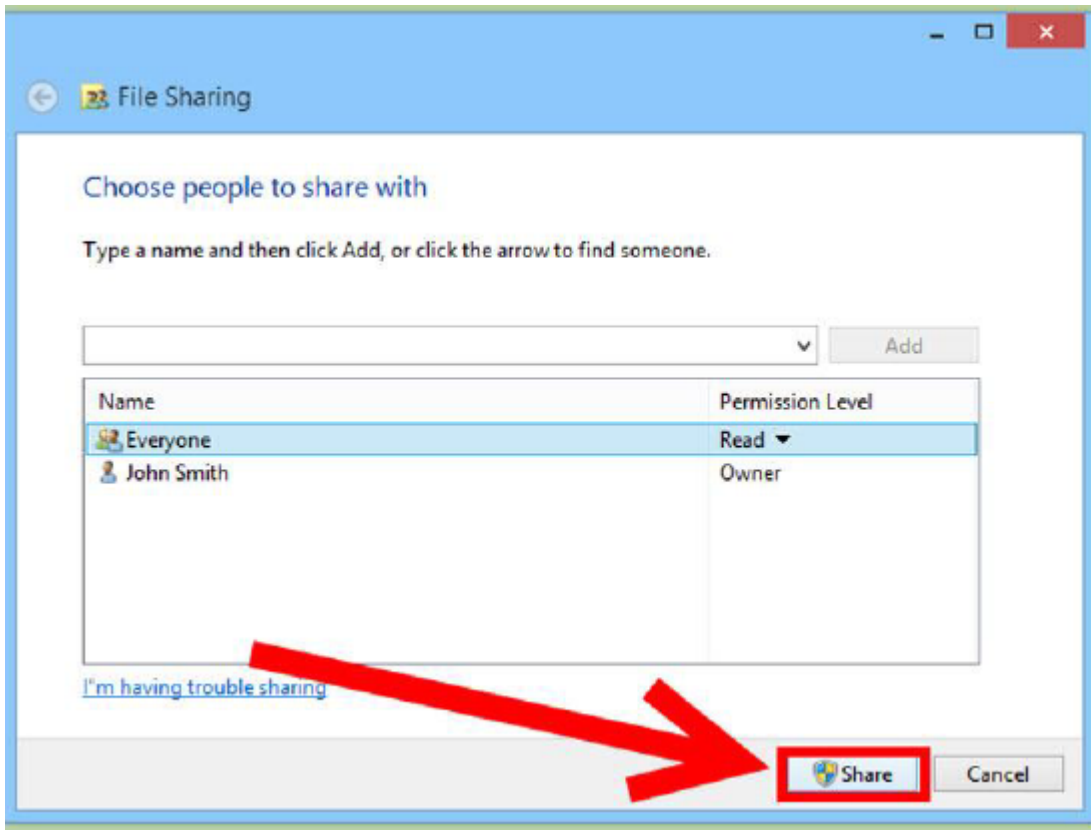
- To share the folder with everyone, click the dropdown menu at the top and select "**Everyone**". Click the Add button.
- To share with specific users, click the dropdown menu and select them or type in the name and click Add.



Set permissions for users on the list. Find a user on the list that you want to change the permissions for. Look in the Permissions Level column, and click the arrow next to the existing permission. Select the new one from the list.

- **Read** - User can see, copy, and open files from the folder, but cannot change files or add new ones.

- **Read/Write** - Besides Read abilities, users can change files and add new files to the shared folder. Files can be deleted by users with Read/Write permissions.
- **Remove** - Removes permissions for this user, and removes them from the list.



Click the Share button. Your permission settings will be saved, and the folder will be available on the network for all allowed users.

Expt.No-16: Use pathping command to find actual path between source to destination with information about network latency/delay & network loss. Also

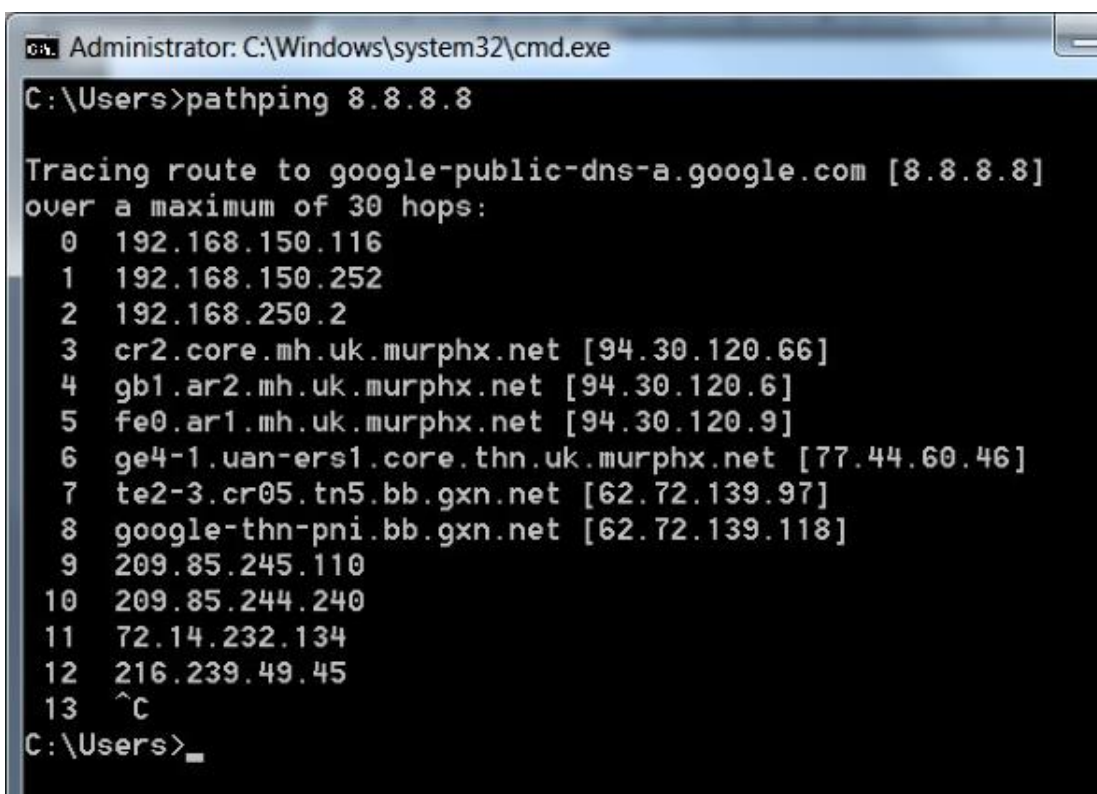
Determine the MAC Address of a Host (PC and Phone).

Pathping usage:

This network utility is a more advanced version of the Ping tool, which performs a ping to each hop along the route to the destination (unlike Ping, which just pings from the originating device to the destination device). It is extremely useful in diagnosing packet loss, and can help with diagnosing slow speed faults.

To PathPing a device, proceed as follows.

1. Open a Windows Command Prompt window.
2. At the command prompt, type, pathping <IP address>, as shown below.



```
Administrator: C:\Windows\system32\cmd.exe
C:\Users>pathping 8.8.8.8

Tracing route to google-public-dns-a.google.com [8.8.8.8]
over a maximum of 30 hops:
 0  192.168.150.116
 1  192.168.150.252
 2  192.168.250.2
 3  cr2.core.mh.uk.murphx.net [94.30.120.66]
 4  gb1.ar2.mh.uk.murphx.net [94.30.120.6]
 5  fe0.ar1.mh.uk.murphx.net [94.30.120.9]
 6  ge4-1.uan-ers1.core.thn.uk.murphx.net [77.44.60.46]
 7  te2-3.cr05.tn5.bb.gxn.net [62.72.139.97]
 8  google-thn-pni.bb.gxn.net [62.72.139.118]
 9  209.85.245.110
10  209.85.244.240
11  72.14.232.134
12  216.239.49.45
13  ^C
C:\Users>
```

Understanding PathPing results

- The advantages of PathPing over Ping and Traceroute are that each node is pinged as the result of a single command, and that the behaviour of nodes is studied over an extended time period, rather than the default ping sample of four messages, or default traceroute single route trace.
- The disadvantage is that it takes a total of 25 seconds per hop to show the PathPing statistics.

Determine the MAC Address of a Host (PC and Phone).

- Follow these instructions to find any computer's MAC address.
 - **Open the command prompt.** Click on the Windows Start menu and search "cmd" or "Command Prompt" in the taskbar.
 - Then type **ipconfig/all** to find the MAC/Physical address.

```
Microsoft Windows [Version 10.0.19044.1526]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ipconfig/all

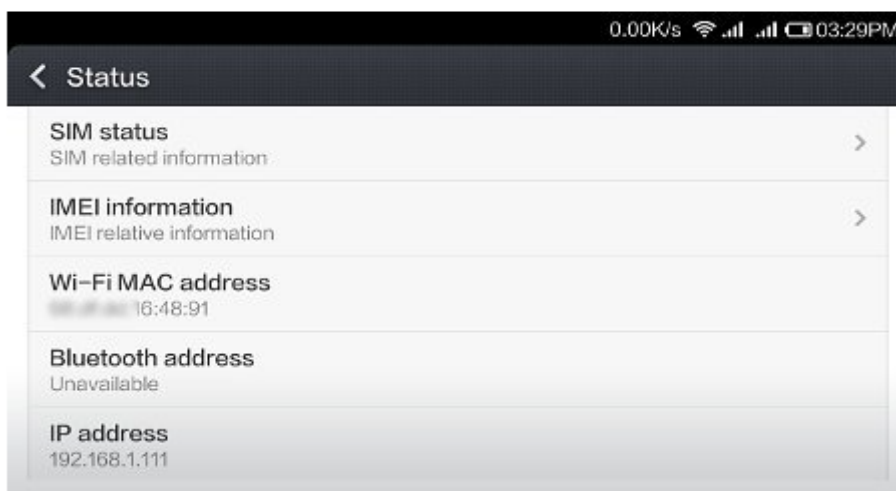
Windows IP Configuration

Host Name . . . . . : Z422
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek PCIe GBE Family Controller
Physical Address. . . . . : 80-EE-73-D4-DA-43
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

- Follow these instructions to find MAC address on any android phone.
 - Open the **Settings** menu.
 - Scroll down and select **About phone**.
 - Select **Status** (or Hardware information).
 - Scroll down to **Wi-Fi MAC address** - this is your device's MAC address.



Expt.No-17: Configure and install a Ethernet switch/Hub (Use simulator if hardware devices are not available)

How to configure Cisco switch in packet tracer?

- In this experiment, we'll see how to configure a Cisco switch. Once you configure your Cisco switch, you can start managing your Cisco switch and maintain its versatility, power and performance.
- Learn how to configure and manage a Cisco Switch in packet tracer with the basic CLI switch commands and configuration steps.

Steps to configure Cisco switch using CLI (Command Line Interface)

Step 1:

- Double click on the switch to open CLI tab.
- Initial command prompt "**Switch>**" appears on the screen.
- Type "**enable**" next to it and press "Enter".
- This will take you into the "**EXEC**" mode, also known as the Global Configuration mode.
- Go into configure mode using configure terminal.
- **Switch#** **configure** **terminal**
Switch(config)#
- Enter the configuration commands one per line.

Step 2: Provide a hostname for the switch to function in a particular network environment

Switch(config)#hostname s1

Now switch configuration changes as follows

S1(config)#

Step 3: Configure an administration password (enable secret password)

S1(config)#enable secret class

Note: This password will have to be given before entering into config mode, once it is configured.

Step 4: Configure the IP address of the switch as 192.168.1.2/24 and it's default gateway IP (192.168.1.1).

```
gptacs(config)#interface Vlan1
```

```
gptacs(config-if)#ip address 192.168.1.2 255.255.255.0
```

```
gptacs(config-if)#ip default-gateway 192.168.1.1
```

```
Switch>enable
Switch#configure
Configuring from terminal, memory, or network [terminal]? terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#enable password cisco
S1(config)#service password-encryption
S1(config)#enable secret class
```

```
S1(config)#banner motd "warning unauthorized person"
S1(config)#line console 0
S1(config-line)#password letmain
S1(config-line)#login
S1(config-line)#exit
S1(config)#line vty 0 4
S1(config-line)#password cisco1
S1(config-line)#login
S1(config-line)#exit
S1(config)#interface vlan1
S1(config-if)#ip address 192.168.1.1 255.255.255.0
S1(config-if)#no shut down
```

Expt.No-18: Create/model a simple Ethernet network using 3 hosts and a switch, Observe traffic behavior on the network and Observe data flow of ARP broadcasts and pings.

Objectives

1. Develop an understanding of the basic functions of Packet Tracer.
2. Model a simple network and observe traffic behaviour on the network.
3. Create a simple Ethernet network using 3 hosts and a switch.
4. Observe data flow of ARP broadcasts and pings.

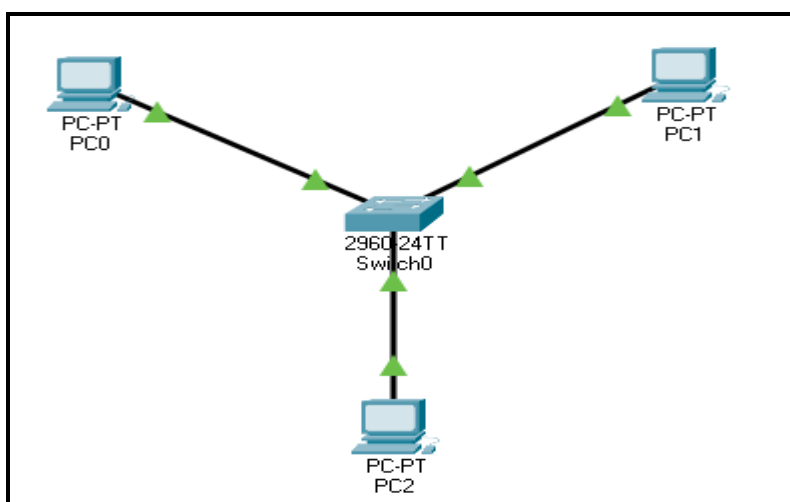
Background / Preparation

Create a logical network diagram with 3 PCs and a switch, connected with straight-through cables.

Step 1: Build the network

The bottom left hand corner of the screen displays the icons that represent device categories or groups, such as Routers, Switches, or End Devices. Moving the cursor over the device categories will show options to the right. To select a device, first select the device category and then select the option that is required.

1. Select **End Devices** from the options in the bottom left-hand corner. Drag and drop 3 **Generic PCs** onto your design area.
2. Select **Switches** from the options in the bottom left-hand corner. Add a **2960** switch to the design area.
3. Select **Connections** from the bottom left-hand corner and choose **Copper Straight-Through** connect each PC to the switch.
 1. PC0 connects to Fast Ethernet 0/1
 2. PC1 connects to Fast Ethernet 0/2
 3. PC2 connects to Fast Ethernet 0/3



Step 2: Configure network devices

1. Select a **PC** and choose the **Desktop** tab.
2. Select **IP Configuration**.
3. Set the IP Address, Subnet Mask, and Default Gateway.

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4. Repeat steps a – c for each PC.

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	192.168.1.11	255.255.255.0	192.168.1.1
PC2	192.168.1.12	255.255.255.0	192.168.1.1

Step 3: Observe traffic flow

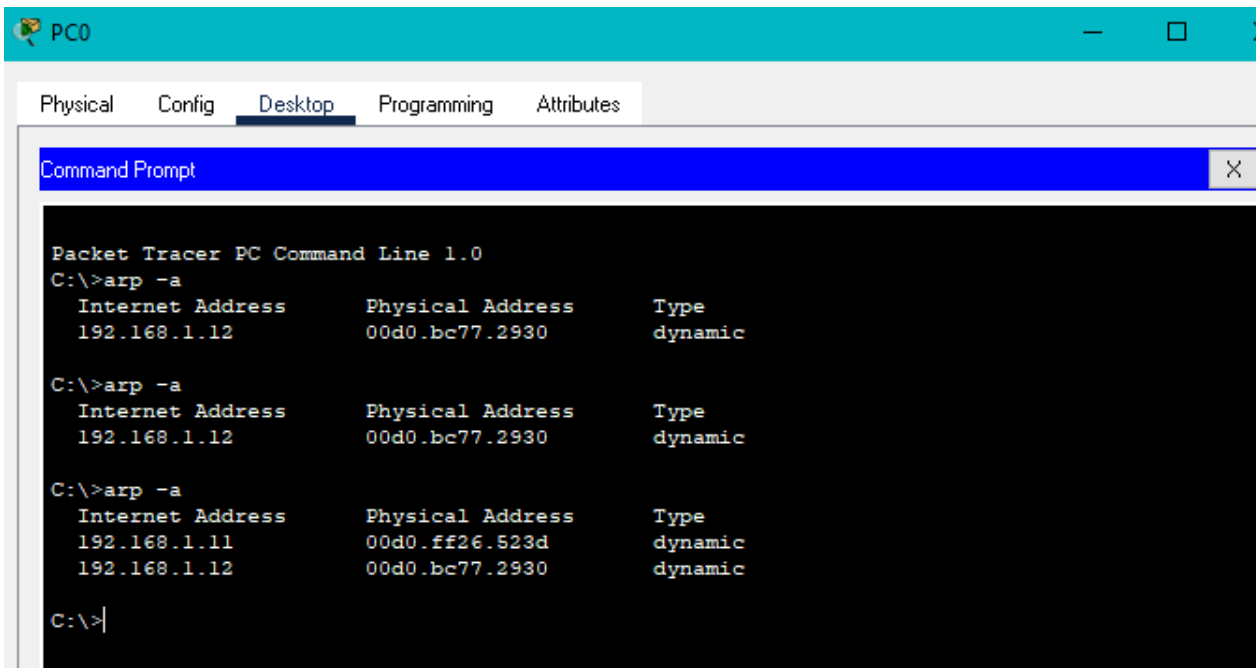
1. Switch to **Simulation Mode** which is in the bottom right-hand corner. The tab has the icon of a stopwatch on it.
2. Select **Edit Filters** and ensure that only ARP and ICMP are selected.
3. Add a Simple PDU by clicking the closed envelope on the right vertical toolbar. Move to PC0 and click to establish the source. Move to PC3 and click to establish the destination.

Notice that two envelopes are now positioned beside PC-A. **One envelop is ICMP, while the other is ARP.** The Event List in the Simulation Panel will identify exactly which envelop represents ICMP and which represents an ARP.

4. Select **Auto Capture / Play** and observe the entire ARP and ICMP communication.

Step 4: View ARP Tables.

1. Select **PC0** and choose the **Desktop** tab.
2. Select the **Command Prompt** and type the command **arp -a**.
Notice that the MAC address for PC2 is in the table.
3. Examine the ARP tables for **PC1** and **PC2**.



The screenshot shows the PC0 Desktop window with the Command Prompt open. The Command Prompt displays the output of the 'arp -a' command, showing the ARP table for PC0. The table lists the Internet Address, Physical Address, and Type for each entry. The first entry is for 192.168.1.12 with Physical Address 00d0.bc77.2930 and Type dynamic. The second entry is for 192.168.1.11 with Physical Address 00d0.ff26.523d and Type dynamic. The third entry is for 192.168.1.12 with Physical Address 00d0.bc77.2930 and Type dynamic.

```
Packet Tracer PC Command Line 1.0
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.12          00d0.bc77.2930       dynamic

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.12          00d0.bc77.2930       dynamic

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.11          00d0.ff26.523d       dynamic
  192.168.1.12          00d0.bc77.2930       dynamic

C:\>
```

Expt.No-19: Build a simple peer-to peer network and verify physical connectivity and Assign various IPv4 addresses to hosts and observe the effects on network communication.

Objectives

- Build a simple peer-to-peer network and verify physical connectivity.
- Assign various IPv4 addresses to hosts and observe the effects on network communication

Background / Scenario

In this experiment, you will build a simple peer-to-peer network using two PCs and an Ethernet crossover cable. You will assign various IPv4 addresses to the hosts and determine the effects on their ability to communicate.

Required Resources

- 2 PCs (Windows 10 with at least one wired Ethernet NIC on each PC)
- 1 Ethernet cross-over cable to connect the PCs



Part 1: Create a Peer-to-Peer Network

Step 1: Connect the PCs to create a peer-to-peer network.

- Obtain an Ethernet crossover cable provided to connect the two PCs.
- Plug one end of the cable into the Ethernet NIC of PC-A. Plug the other end of the cable into the Ethernet NIC of PC-B. As you insert the cable, you should hear a click which indicates that the cable connector is properly inserted into the port.

Step 2: Verify physical connectivity.

- After the Ethernet crossover cable is connected to both PCs, take a close look at each Ethernet port. A link light (usually green or amber) indicates that physical connectivity has been established between the two NICs. Try unplugging the cable from one PC then reconnecting it to verify that the light goes off then back on.
- On PC-A, right-click **Start** and select **Network Connections**.
- If there was a problem connecting the network cable, Ethernet0 will read **Network cable unplugged** and will have a red X in the icon. If so, troubleshoot by repeating Steps 1 and 2.

Step 3: Configure IPv4 Settings on PC-A and PC-B.

- Configure manual IPv4 addressing on PC-A and PC-B so that they can communicate using TCP/IP. On PC-A, right click the **Ethernet0** and click **Properties**.
- In the Ethernet0 Properties window select **Internet Protocol Version 4 (TCP/IPv4)** and click the **Properties** button.
- Select the **Use the following IPv4 address** and enter the IPv4 address **192.168.1.1** and subnet mask **255.255.255.0**. Click **OK** > **Close** to exit the Ethernet0 Properties window.

- d. Repeat steps 3a – 3c for PC-B using an IPv4 address **192.168.1.2** and a subnet mask **255.255.255.0**.

Step 4: Verify IPv4 connectivity between the two PCs.

Note: To test TCP/IP connectivity between the PCs, Windows Firewall must be disabled temporarily on both PCs. Windows Firewall should be re-enabled after the tests have been completed. To access Windows Firewall, click **Settings > Network & Internet > Ethernet > Windows Firewall > Turn Windows Firewall on or off**.

- a. Now that the two PCs are physically connected and configured correctly with IPv4 addresses, we need to make sure they can communicate with each other. The **ping** command is a simple way to accomplish this task.
- b. Navigate to the Command Prompt on PC-A. In a Command Prompt on PC-A, ping **192.168.1.2**. A successful **ping** will verify network connectivity and PC-A can communicate with PC-B.
- C:\Users\Student> **ping 192.168.1.2**

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=4ms TTL=64

Reply from 192.168.1.2: bytes=32 time=1ms TTL=64

Reply from 192.168.1.2: bytes=32 time=1ms TTL=64

Reply from 192.168.1.2: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.1.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 4ms, Average = 1ms

- c. Repeat this procedure and **ping 192.168.1.1** from PC-B.
- d. Close the command prompt on both PCs.

Part 2: Modify IPv4 Addresses

Step 1: Change IPv4 address for PC-B.

- a. On PC-B, right-click on **Start** and select **Network Connections**, and right-click the **Ethernet0** icon. Choose **Properties** from the pull-down menu.
- b. Select Internet Protocol Version 4 (TCP/IPv4). Click **Properties**.
- c. Change the logical IPv4 address for PC-B from 192.168.1.2 to **192.168.2.2** and leave the subnet mask set to 255.255.255.0.
- d. Click **OK** which will close the **Internet Protocol Version 4 (TCP/IPv4)** window. Click the **Close** button to exit the **Ethernet0 Properties** window.

Step 2: Test network connectivity between the 2 PCs.

- a. From PC-B, right-click **Start**, and select **Command Prompt**.
- b. At the command prompt, type **ping 192.168.2.2** and press **Enter**.

Questions:

Was it successful? Explain.

The ping was unsuccessful because PC-A and PC-B are no longer in the same subnet. Without a router, these PCs cannot communicate with each other.

What type of networking device would allow the PCs to communicate even though they are on different networks?

A router is needed to allow the PCs to communicate.

Step 3: Change IPv4 address for PC-A.

- a. On PC-A, right-click **Start** and select **Network Connections**, and right-click the **Ethernet0** icon. Choose **Properties** from the pull-down menu.
- b. Select Internet Protocol Version 4 (TCP/IP). Click Properties.
- c. Change the logical IPv4 address for PC-A from 192.168.1.1 to 192.168.2.99 and leave the subnet mask set to 255.255.255.0. Click **OK**, to close the **Internet Protocol Version 4 (TCP/IP)** window.
Click **Close** to exit the **Ethernet0 Properties** window.

The two PCs are still on the same physical Ethernet network.

Question:

Are they on the same logical IPv4 network now?

Yes, they are in the same physical and logical network.

Step 4: Test network connectivity between the 2 PCs.

1. On PC-B, repeat step 4b to access the Windows command prompt.
2. At the command prompt, **ping 192.168.2.99**.

Question:

Was it successful? Explain.

The ping was successful because PC-A and PC-B are in the same subnet.

Step 5: (Optional – Use only if the Firewall was originally ENABLED) Re-enable the firewall.

To ensure that the PC is protected from unauthorized access, re-enable the Windows Firewall.

To access Windows Firewall, click **Settings > Network & Internet > Ethernet > Windows Firewall > Turn Windows Firewall on or off**.

Expt.No-20: Configure IP addresses of a network (real or simulated) and ping across to test and troubleshoot.

Objectives

1. Develop an understanding of the basic functions of Packet Tracer.
2. Model a simple network and observe traffic behaviour on the network.
3. Create a simple Ethernet network using 2 hosts and a switch.
4. Observe data flow using pings.

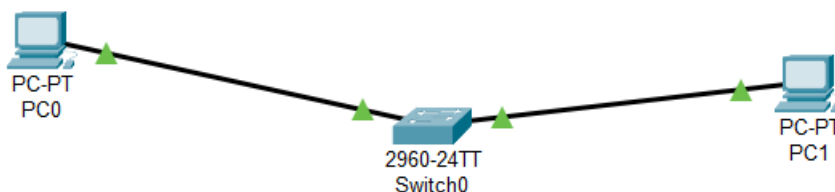
Background / Preparation

Create a logical network diagram with 2 PCs and a switch, connected with straight-through cables.

Step 1: Build the network

The bottom left hand corner of the screen displays the icons that represent device categories or groups, such as Routers, Switches, or End Devices. Moving the cursor over the device categories will show options to the right. To select a device, first select the device category and then select the option that is required.

1. Select **End Devices** from the options in the bottom left-hand corner. Drag and drop 3 **Generic PCs** onto your design area.
2. Select **Switches** from the options in the bottom left-hand corner. Add a **2960** switch to the design area.
3. Select **Connections** from the bottom left-hand corner and choose **Copper Straight-Through** connect each PC to the switch.
 1. PC0 connects to Fast Ethernet 0/1
 2. PC1 connects to Fast Ethernet 0/2



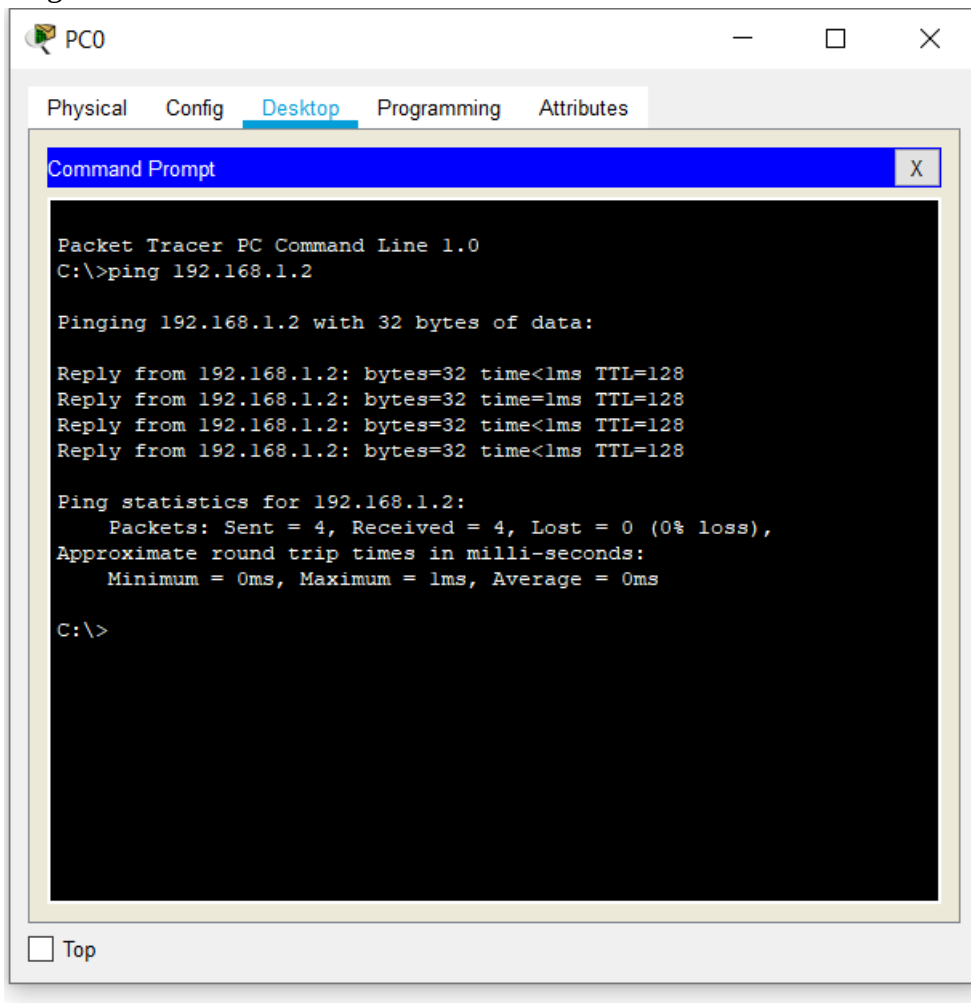
Step 2: Configure network devices

1. Select a **PC** and choose the **Desktop** tab.
2. Select **IP Configuration**.
3. Set the IP Address, Subnet Mask, and Default Gateway.
4. Repeat steps a – c for each PC.

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0

Step 3: Observe Result

1. Ping from PC0 to PC1



Expt.No-21: Subnetting of a network (either using real network or in Simulator).

Objectives

1. Develop an understanding of the basic functions of Packet Tracer.
2. Model a simple network and observe traffic behaviour on the network.
3. Create a subnet of network.
4. Observe status in status bar after transmitting PDU and pings.

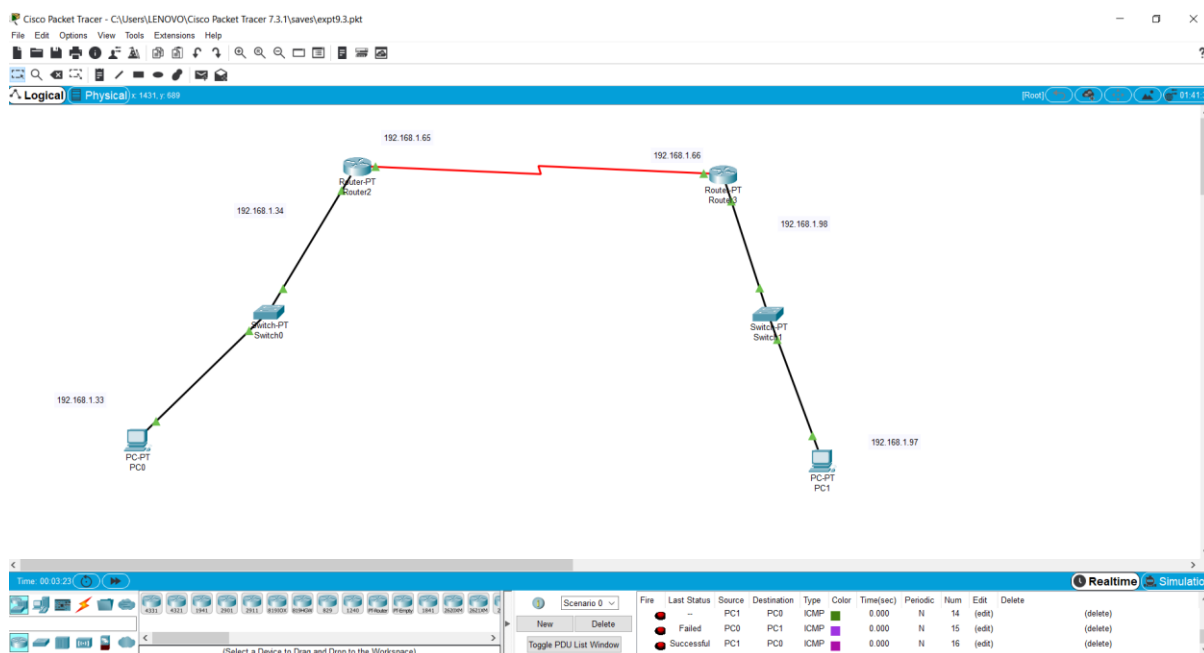
Background / Preparation

Create a logical network diagram with 2 PCs, 2 switch and two routers connected with straight-through cables.

Step 1: Build the network

The bottom left hand corner of the screen displays the icons that represent device categories or groups, such as Routers, Switches, or End Devices. Moving the cursor over the device categories will show options to the right. To select a device, first select the device category and then select the option that is required.

4. Select **End Devices** from the options in the bottom left-hand corner. Drag and drop 2 **Generic PCs** onto your design area.
5. Select **Switches** from the options in the bottom left-hand corner. Add two **PT switch** to the design area.
6. Select **Connections** from the bottom left-hand corner and choose **Copper Straight-Through** connect each PC to the switch.
 1. PC0 connects to Fast Ethernet 0/1
 2. PC1 connects to Fast Ethernet 0/2



Step 2: Configure network devices

1. Select a **PC** and Routers choose the **Desktop** tab.
2. Select **IP Configuration**.
3. Set the IP Address, Subnet Mask, and Default Gateway.
4. Repeat steps a – c for each PC.
5. Static routing in both the router3 192.168.1.32/27 via 192.168.1.65
6. Static routing in both the router2 192.168.1.96/27 via 192.168.1.66

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.33	255.255.224.0	192.168.1.34
PC1	192.168.1.97	255.255.224.0	192.168.1.98
Router0 Inerface Fa0/0	192.168.1.34	255.255.224.0	192.168.1.1
Router0 Inerface Se2/0	192.168.1.65	255.255.224.0	
Router1 Inerface Se3/0	192.168.1.66	255.255.224.0	
Router0 Inerface Fa0/0	192.168.1.98	255.255.224.0	

Step 3: Observe Result

1. Ping from PC1 to PC0

 Realtime  Simulation										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	--	PC1	PC0	ICMP		0.000	N	14	(edit)	(delete)
	Failed	PC0	PC1	ICMP		0.000	N	15	(edit)	(delete)
	Successful	PC1	PC0	ICMP		0.000	N	16	(edit)	(delete)

Expt.No-22: Connect to web server using simulator, Observe how packets are sent across the Internet using IP addresses.

Objectives

1. Develop an understanding of the basic functions of Packet Tracer.
2. Model a simple network and observe traffic behaviour on the network.
3. Create a simple Ethernet network using 1 hosts , a switch and 2 server
4. Observe the result in web browser at PC0.

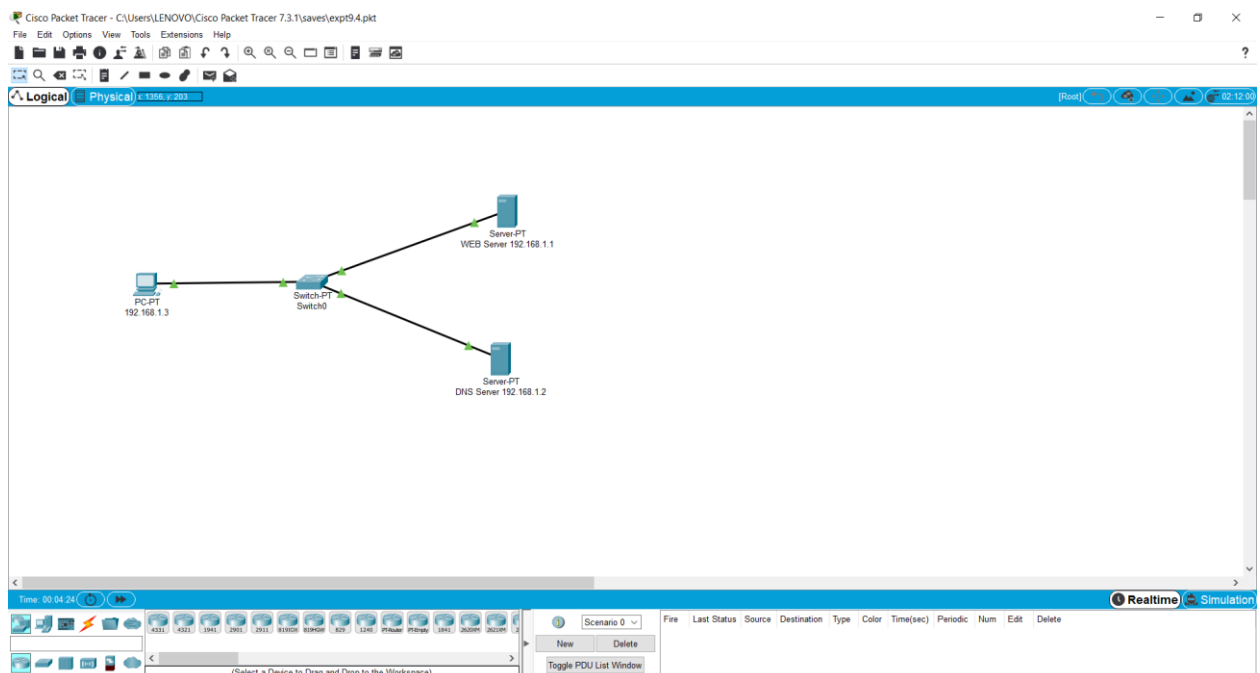
Background / Preparation

Create a logical network diagram with 1 PC , switch and 2 server, connected with straight-through cables.

Step 1: Build the network

The bottom left hand corner of the screen displays the icons that represent device categories or groups, such as Routers, Switches, or End Devices. Moving the cursor over the device categories will show options to the right. To select a device, first select the device category and then select the option that is required.

1. Select **End Devices** from the options in the bottom left-hand corner. Drag and drop 3 **Generic PCs** onto your design area.
2. Select **Switches** from the options in the bottom left-hand corner. Add a **PT** switch to the design area.
3. Select **Server** from the options in the bottom left-hand corner. Add a **Server PT** to the design area
4. Select **Connections** from the bottom left-hand corner and choose **Copper Straight-Through** connect each PC to the switch.
 1. PC0 connects to Fast Ethernet 0/1
 2. WEB SERVER to Fa0/0
 3. DNS Server to Fa0/0

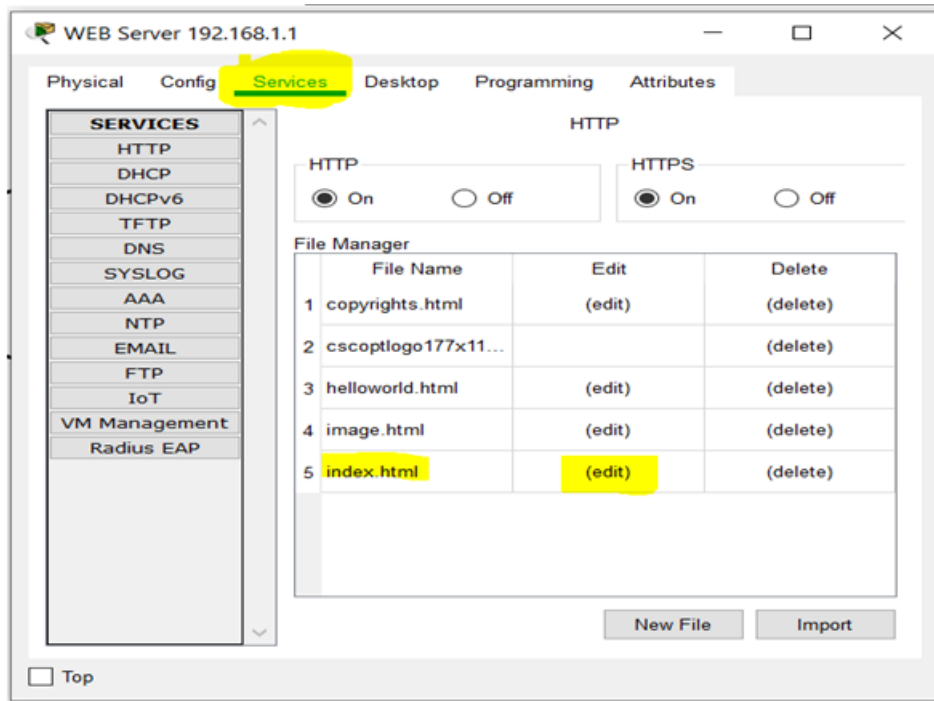


Step 2: Configure network devices

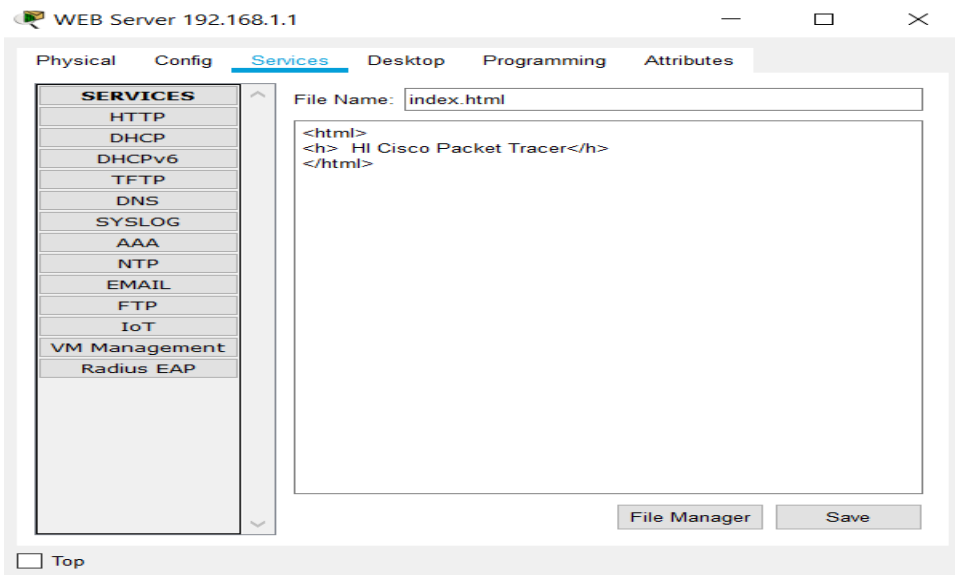
1. Select a **PC** and Routers choose the **Desktop** tab.
2. Select **IP Configuration**.
3. Set the IP Address, Subnet Mask, and Default Gateway.
4. Repeat steps a – c for each PC.
5. Static routing in both the router3 192.168.1.32/27 via 192.168.1.65
6. Static routing in both the router2 192.168.1.96/27 via 192.168.1.66

Device	IP Address	Subnet Mask	DNS Server
PC0	192.168.1.3	255.255.255.0	192.168.1.2
WEB Sever	192.168.1.1	255.255.255.0	192.168.1.2
DNS Server	192.168.1.2	255.255.255.0	192.168.1.2

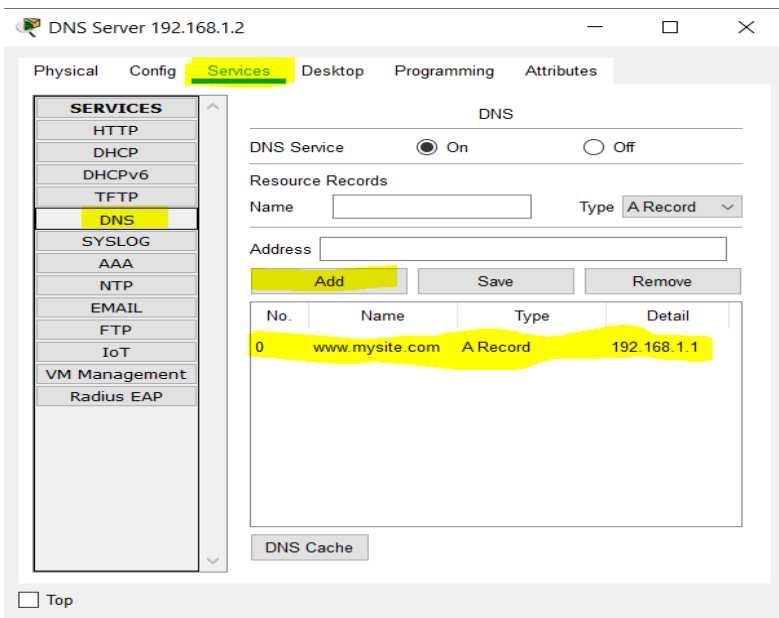
At WEB Server:



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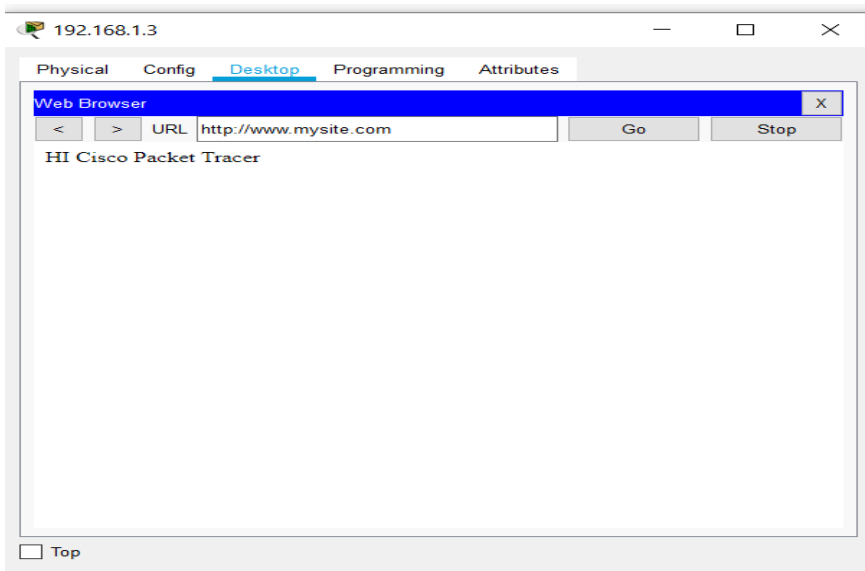
At DNS Server



Step 3: Observe Result

1. Open Web Browser PC0

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Expt.No-23: Implement simple static routing

Objectives

1. Develop an understanding of the basic functions of Packet Tracer.
2. Model a simple network and observe traffic behaviour on the network.
3. Create a subnet of network.
4. Observe status in status bar after transmitting PDU and pings.

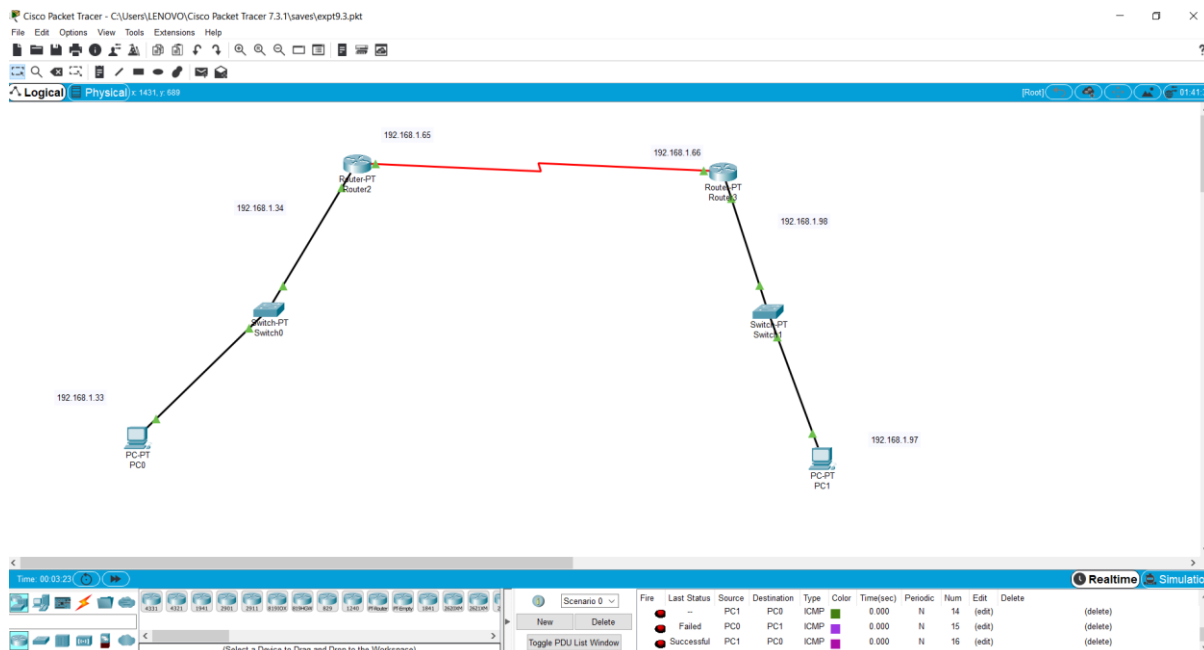
Background / Preparation

Create a logical network diagram with 2 PCs, 2 switch and two routers connected with straight-through cables.

Step 1: Build the network

The bottom left hand corner of the screen displays the icons that represent device categories or groups, such as Routers, Switches, or End Devices. Moving the cursor over the device categories will show options to the right. To select a device, first select the device category and then select the option that is required.

2. Select **End Devices** from the options in the bottom left-hand corner. Drag and drop 2 **Generic PCs** onto your design area.
3. Select **Switches** from the options in the bottom left-hand corner. Add two **PT switch** to the design area.
4. Select **Connections** from the bottom left-hand corner and choose **Copper Straight-Through** connect each PC to the switch.
 1. PC0 connects to Fast Ethernet 0/1
 2. PC1 connects to Fast Ethernet 0/2



Step 2: Configure network devices

7. Select a **PC** and Routers choose the **Desktop** tab.

8. Select **IP Configuration**.
9. Set the IP Address, Subnet Mask, and Default Gateway.
10. Repeat steps a – c for each PC.
11. Static routing in both the router3 192.168.1.32/27 via 192.168.1.65
12. Static routing in both the router2 192.168.1.96/27 via 192.168.1.66

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.33	255.255.224.0	192.168.1.34
PC1	192.168.1.97	255.255.224.0	192.168.1.98
Router0 Interface Fa0/0	192.168.1.34	255.255.224.0	192.168.1.1
Router0 Interface Se2/0	192.168.1.65	255.255.224.0	
Router1 Interface Se3/0	192.168.1.66	255.255.224.0	
Router0 Interface Fa0/0	192.168.1.98	255.255.224.0	

Step 3: Observe Result

5. Ping from PC1 to PC0

Realtime Simulation										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	--	PC1	PC0	ICMP		0.000	N	14	(edit)	(delete)
	Failed	PC0	PC1	ICMP		0.000	N	15	(edit)	(delete)
	Successful	PC1	PC0	ICMP		0.000	N	16	(edit)	(delete)

Expt.No-24: Troubleshooting of IP Addressing a) Wrong address b) incorrect subnet mask.

Objectives

- 1.Develop an understanding of the basic functions of Packet Tracer.
- 2.Model a simple network and observe traffic behaviour on the network.
- 3.Create a simple Ethernet network using 2 hosts and a switch.
- 4.Observe data flow using pings.

Background / Preparation

Create a logical network diagram with 2 PCs and a switch, connected with straight-through cables.

Step 1: Build the network

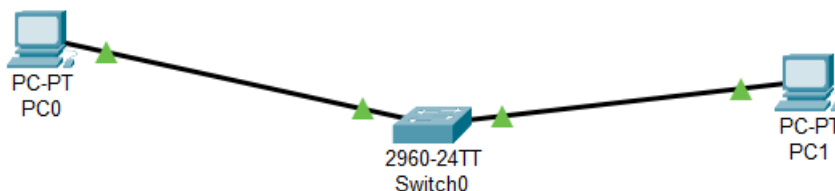
The bottom left hand corner of the screen displays the icons that represent device categories or groups, such as Routers, Switches, or End Devices. Moving the cursor over the device categories will show options to the right. To select a device, first select the device category and then select the option that is required.

1.Select **End Devices** from the options in the bottom left-hand corner. Drag and drop 3 **Generic PCs** onto your design area.

2.Select **Switches** from the options in the bottom left-hand corner. Add a **2960** switch to the design area.

3.Select **Connections** from the bottom left-hand corner and choose **Copper Straight-Through** connect each PC to the switch.

- 1.PC0 connects to Fast Ethernet 0/1
- 2.PC1 connects to Fast Ethernet 0/2



Step 2: Configure network devices

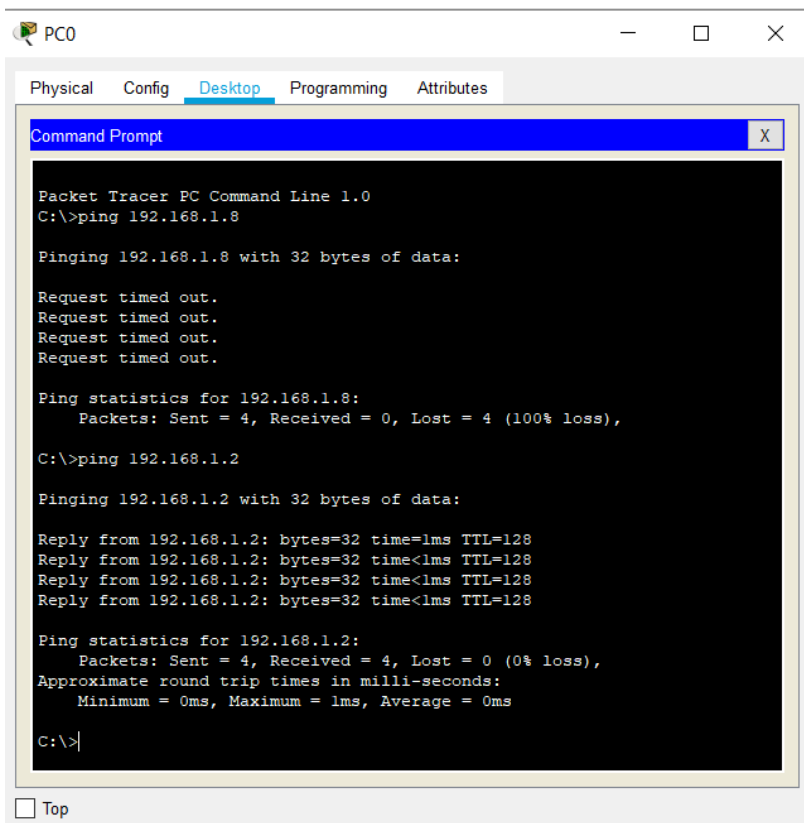
1. Select a **PC** and choose the **Desktop** tab.
2. Select **IP Configuration**.
3. Set the IP Address, Subnet Mask, and Default Gateway.
4. Repeat steps a – c for each PC.

Device	IP Address	Subnet Mask
--------	------------	-------------

PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0

Step 3: Observe Result

1. Ping from PC0 to PC2 with wrong IP address. Find the troubleshoot problem then change to correct IP address
2. ipconfig PC2 with wrong mask. Find the troubleshoot problem then change to mask



The screenshot shows a Packet Tracer PC0 window with the 'Desktop' tab selected. A Command Prompt window is open, displaying the following text:

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.8

Pinging 192.168.1.8 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.1.2

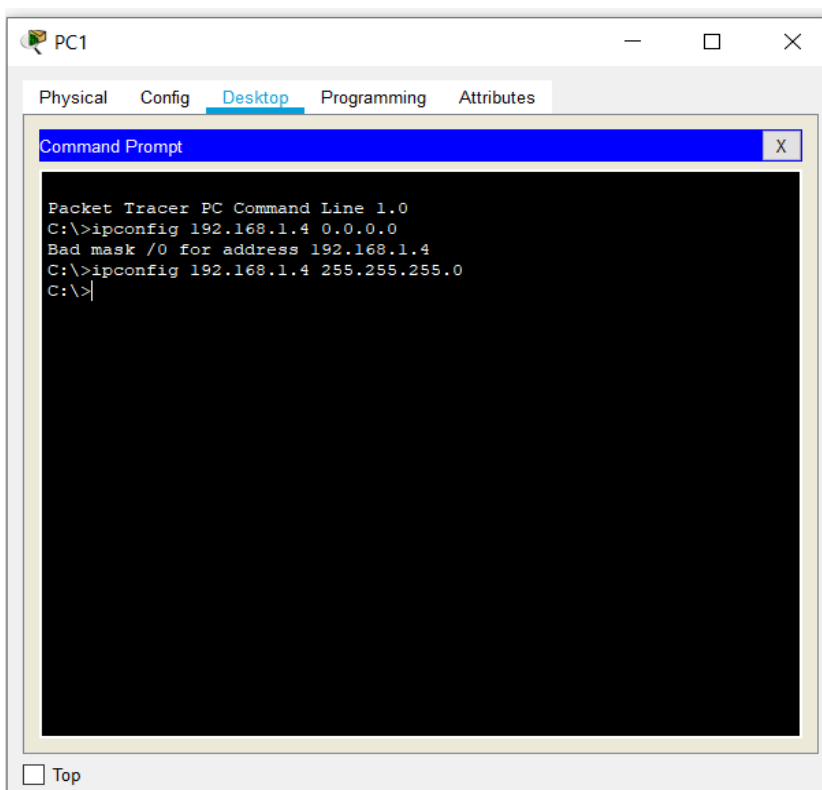
Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

At the bottom left of the Command Prompt window, there is a checkbox labeled 'Top' which is currently unchecked.



Expt.No-25: . Configure and test DHCP on a wireless router (real or simulated)

Objectives

- 1.Develop an understanding of the basic functions of Packet Tracer.
- 2.Model a simple network and observe traffic behaviour on the network.
- 3.Create a simple Ethernet network using 3 hosts and a wireless router.
- 4.Observe data flow using pings.

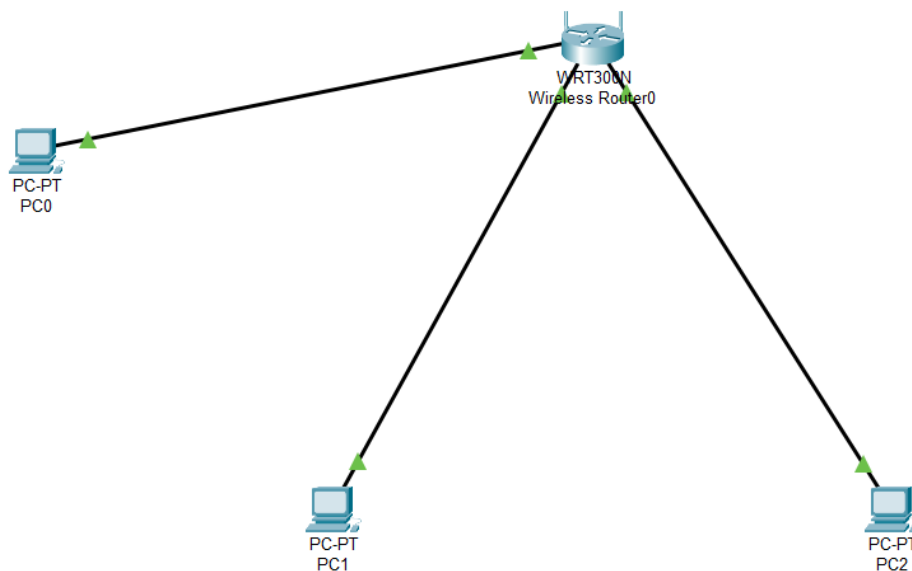
Background / Preparation

Create a logical network diagram with 3 PCs and a wireless router., connected with straight-through cables.

Step 1: Build the network

The bottom left hand corner of the screen displays the icons that represent device categories or groups, such as Routers, Switches, or End Devices. Moving the cursor over the device categories will show options to the right. To select a device, first select the device category and then select the option that is required.

- 1.Select **End Devices** from the options in the bottom left-hand corner. Drag and drop 3 **Generic PCs** onto your design area.
- 2.Select wireless devices. from the options in the bottom left-hand corner. Add WRT300N router to the design area.
3. Select **Connections** from the bottom left-hand corner and choose **Copper Straight-Through** connect each PC to the switch.
 - 1.PC0 connects to Fast Ethernet 0/1
 - 2.PC1 connects to Fast Ethernet 0/2
 - 3.PC2 connects to Fast Ethernet 0/3

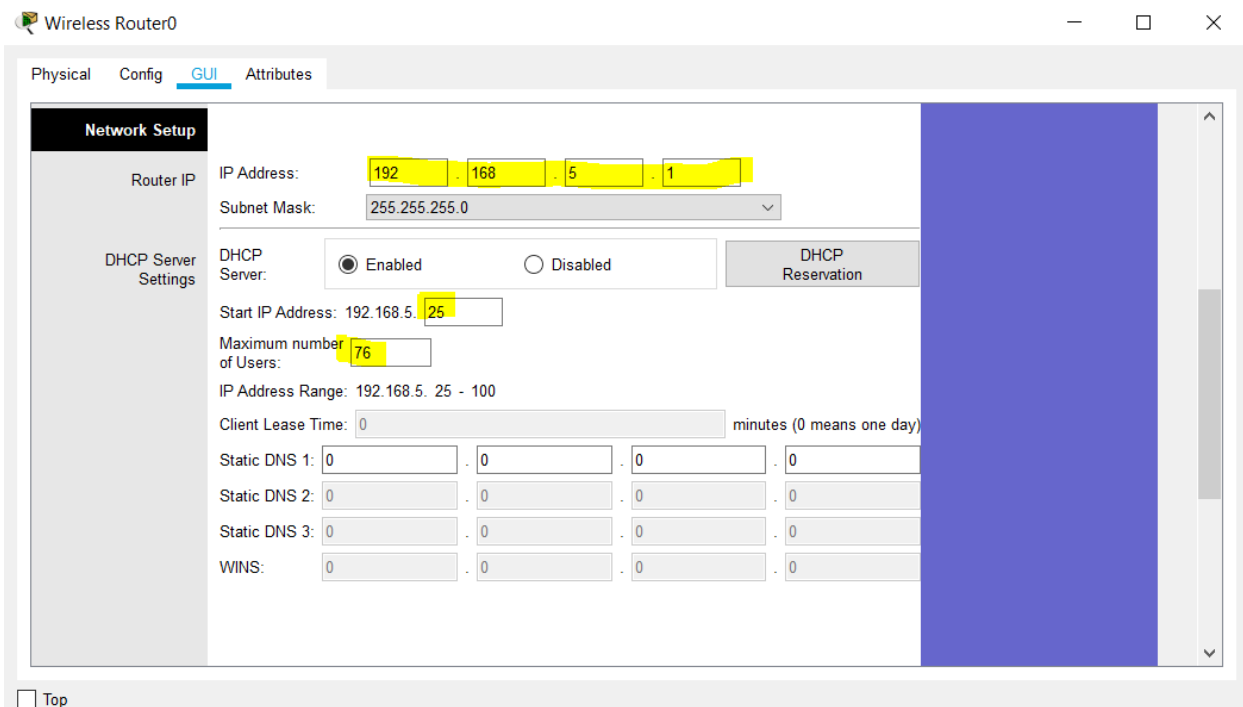
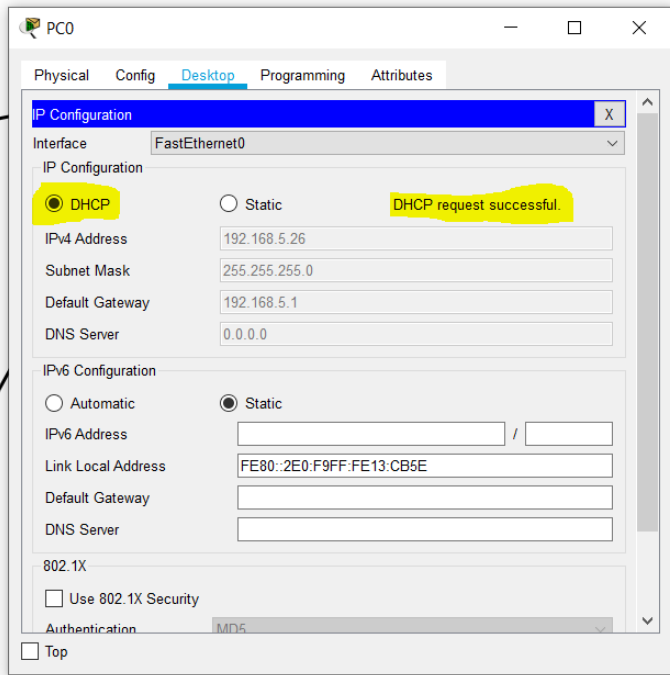


Step 2: Configure network devices

1. Select a **PC** and choose the **Desktop** tab.
2. Select **IP Configuration**.

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3. Select DHCP option.
4. Repeat steps a – c for each PC.
5. Perform GUI settings in Wireless router



Step 3: Observe Result

1. Ping from PC0 to PC2
2. Ping from PC1 to PC0

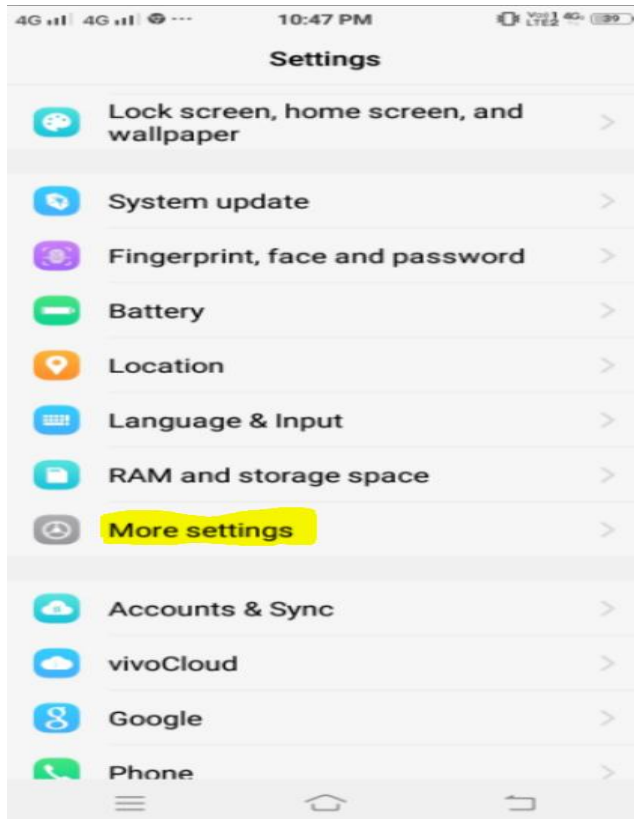
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Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC0	ICMP		0.000	N	1	(edit)	(delete)

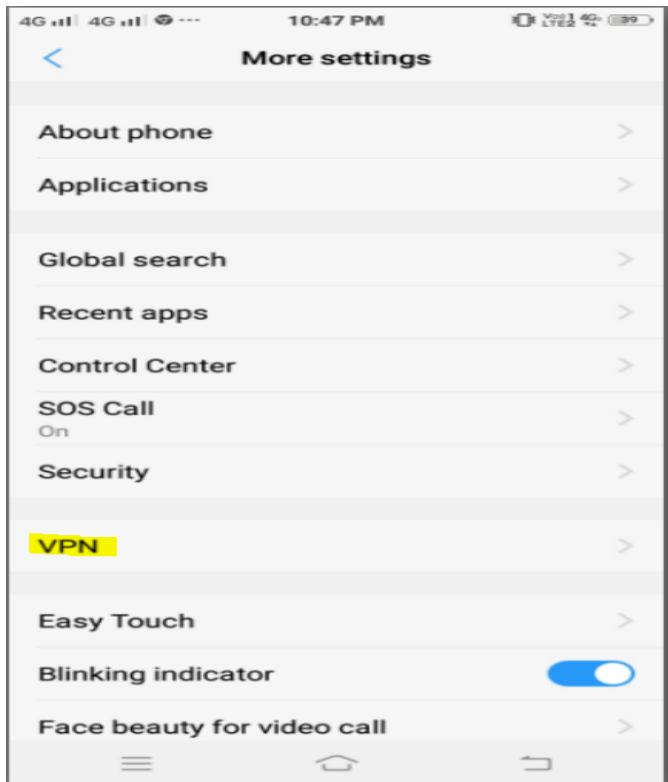
Expt.No-26: Setup, configure and test VPN in your smartphone

To set VPN network

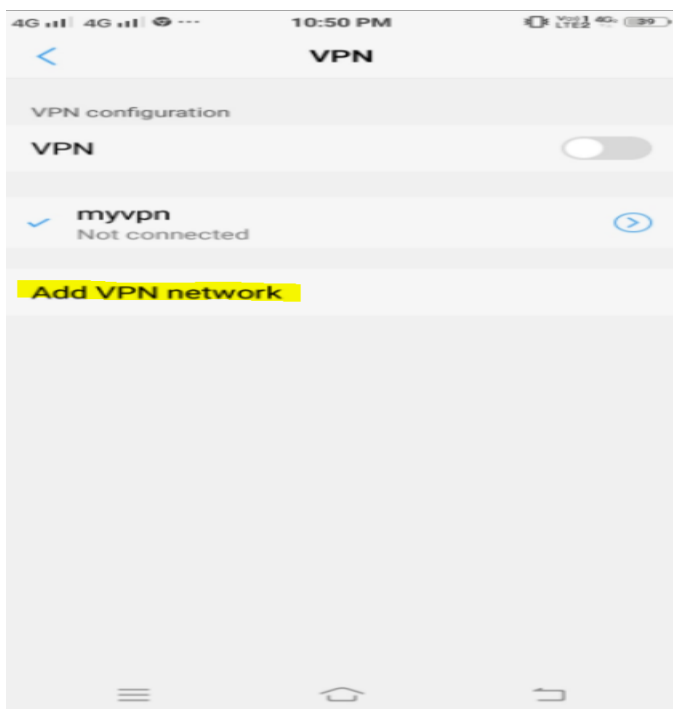
- Open the setting window in the mobile phone
- Select more settings



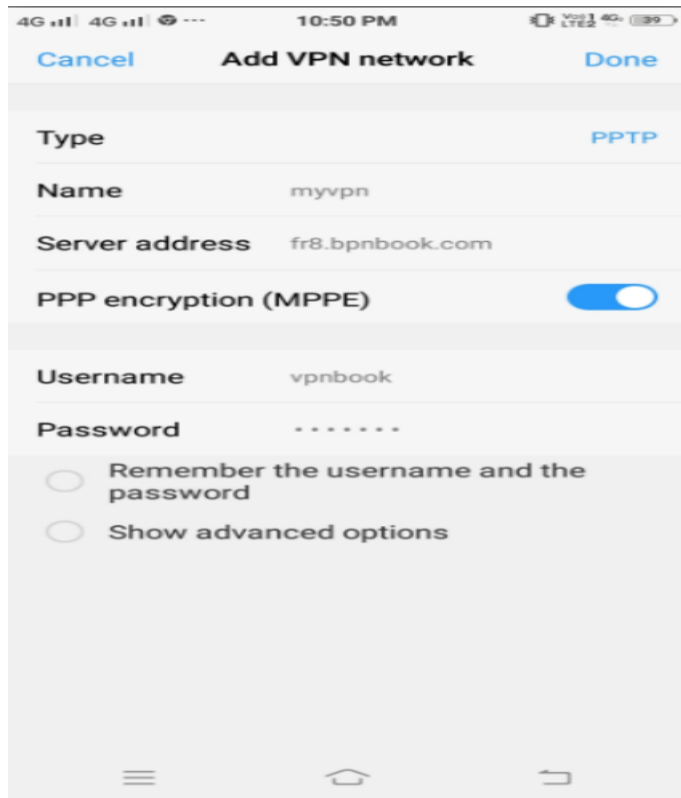
- Select VPN option



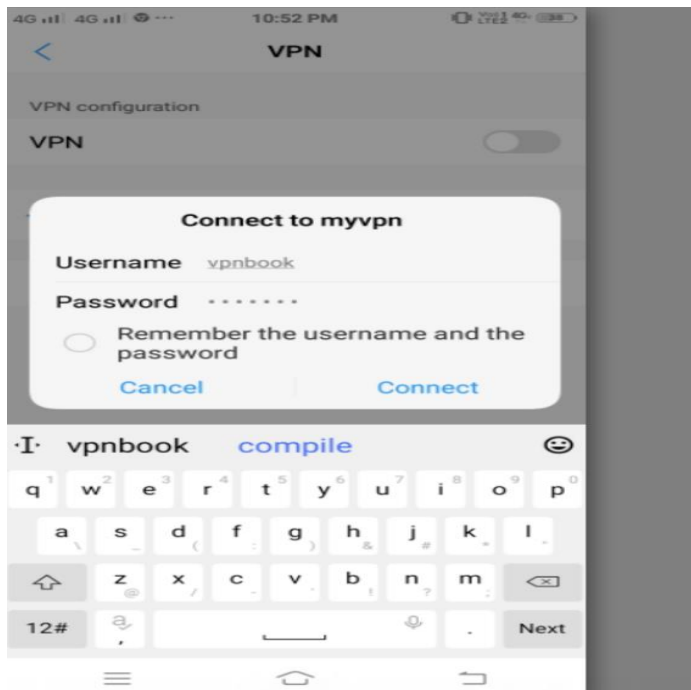
- Then select add VPN network



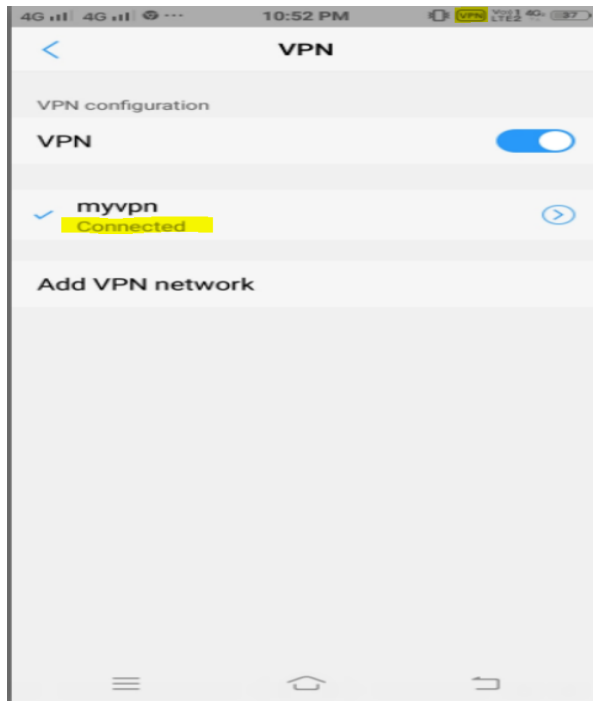
- Enter the following fields
 1. Name :myvpn
 2. Sever address:fr8.vpnbook.com
 3. Username :vpnbook
 4. Password:fx79vrh



- Then click on done the following window appears then enter appropriate username and password



- Then finally VPN network will be created



Expt.No-27: Create a client – server model in simulator and observe the client interaction between the server and PC using packet tracer

Objectives

1. Develop an understanding of the basic functions of Packet Tracer.
2. Model a simple network and observe traffic behaviour on the network.
3. Create a simple Ethernet network using 4 hosts as client and a server.
4. Observe data flow using pings.

Background / Preparation

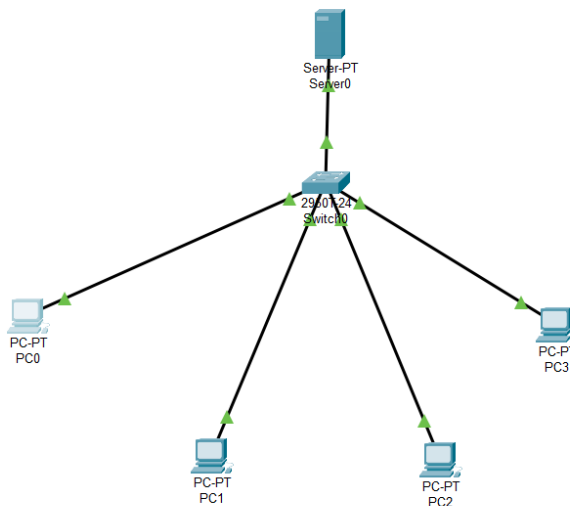
Create a logical network diagram with 4 PCs and a Server, connected with straight-through cables.

Step 1: Build the network

The bottom left hand corner of the screen displays the icons that represent device categories or groups, such as Routers, Switches, or End Devices. Moving the cursor over the device categories will show options to the right. To select a device, first select the device category and then select the option that is required.

1. Select **End Devices** from the options in the bottom left-hand corner. Drag and drop 3 **Generic PCs** onto your design area.
2. Select **Server** from the options in the bottom left-hand corner. Add a **Server PT** to the design area.
3. Select **Connections** from the bottom left-hand corner and choose **Copper Straight-Through** connect each PC to the switch.

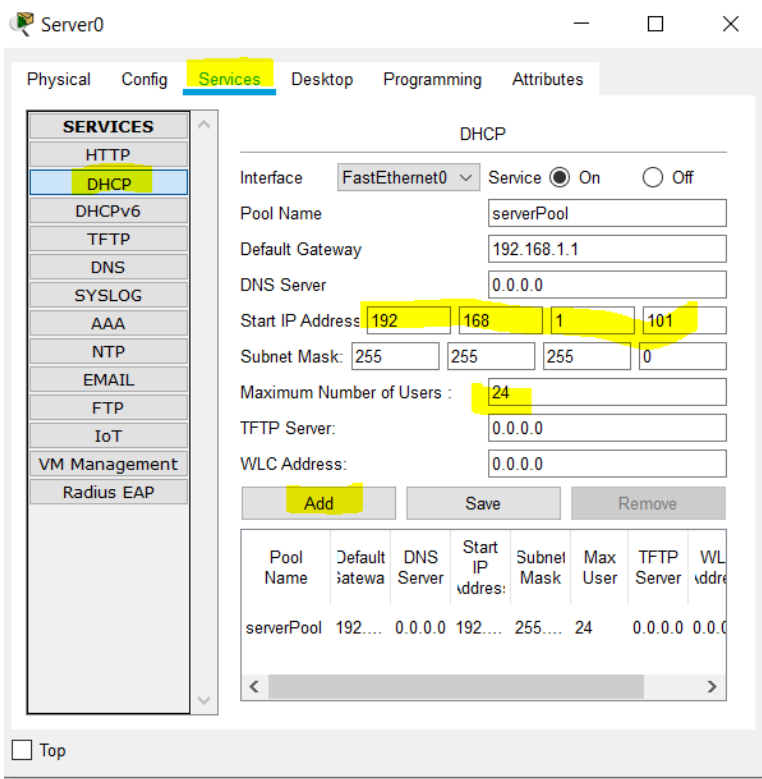
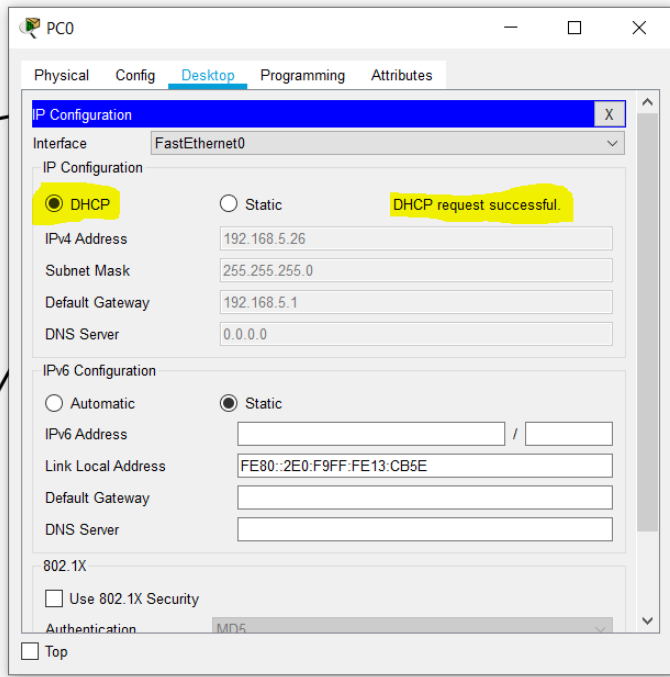
1. PC0 connects to Fast Ethernet 0/1
2. PC1 connects to Fast Ethernet 0/2
3. PC2 connects to Fast Ethernet 0/3
4. PC3 connects to Fast Ethernet 0/4



Step 2: Configure network devices

1. Select a **PC** and choose the **Desktop** tab.
2. Select **IP Configuration**.
3. Select DHCP option.
4. Repeat steps a – c for each PC.
5. Perform services change in Server

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Step 3: Observe Result

1. Ping from PC0 to PC2
3. Ping from PC1 to PC0

RealtimeSimulation										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC0	ICMP		0.000	N	1	(edit)	(delete)

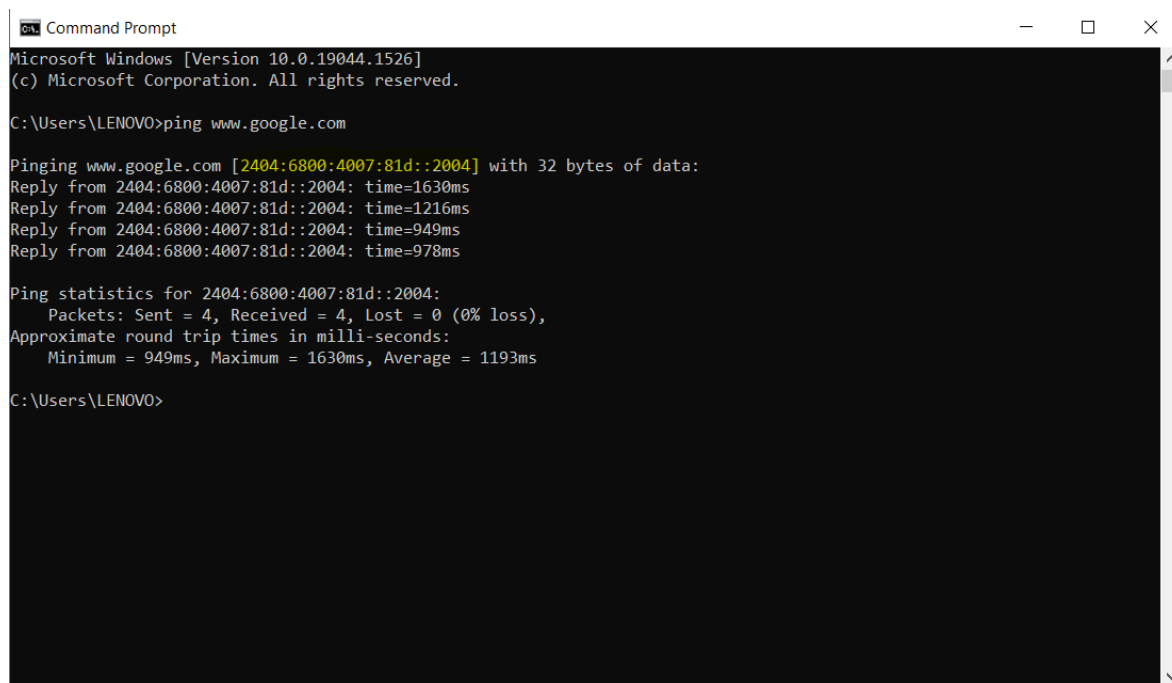
Expt.No-28: Observe DNS Name Resolution

- a) Observe the conversion of a URL to an IP address.
- b) Observe DNS lookup using the nslookup command.

a)To obtain IP address from URL

To Ping a device, proceed as follows.

1. Open a Windows Command Prompt window.
2. At the command prompt, type, ping DNS of a site , as shown below.



```
Command Prompt
Microsoft Windows [Version 10.0.19044.1526]
(c) Microsoft Corporation. All rights reserved.

C:\Users\LENOVO>ping www.google.com

Pinging www.google.com [2404:6800:4007:81d::2004] with 32 bytes of data:
Reply from 2404:6800:4007:81d::2004: time=1630ms
Reply from 2404:6800:4007:81d::2004: time=1216ms
Reply from 2404:6800:4007:81d::2004: time=949ms
Reply from 2404:6800:4007:81d::2004: time=978ms

Ping statistics for 2404:6800:4007:81d::2004:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 949ms, Maximum = 1630ms, Average = 1193ms

C:\Users\LENOVO>
```

Understanding results:

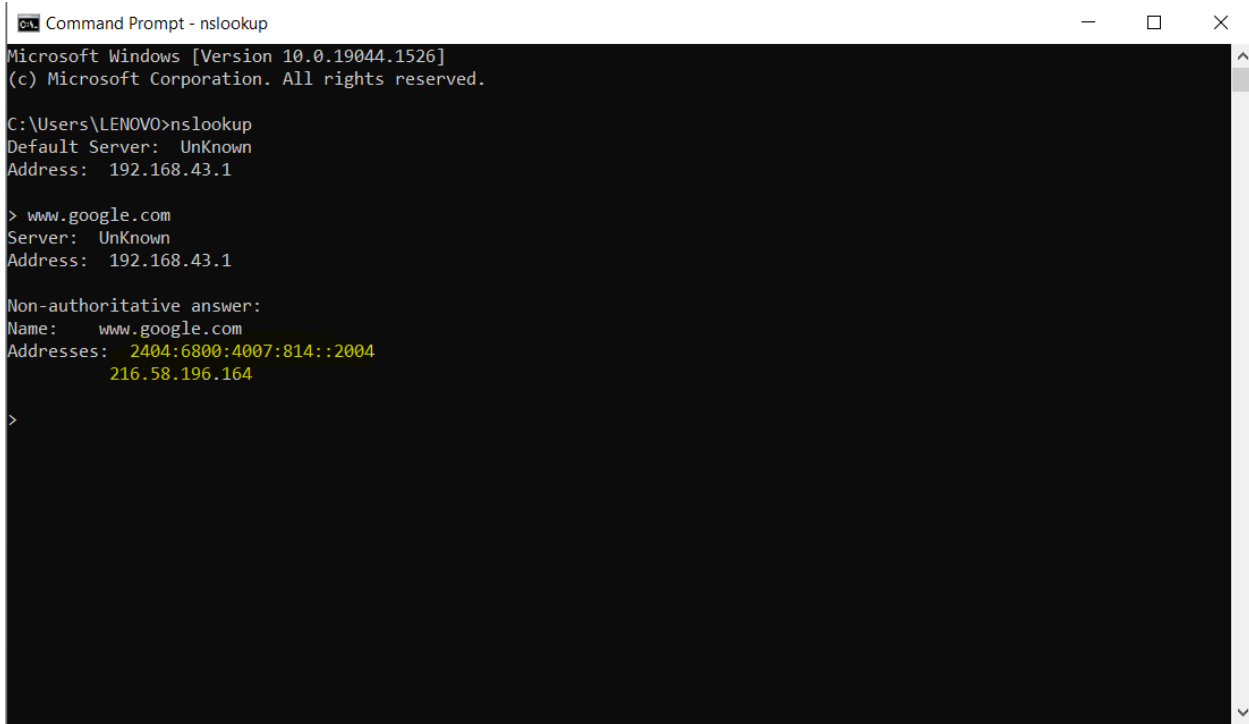
Observe the above result pinging first line indicates IP address of www.google.com

- b) Observe DNS lookup using the nslookup command

To Ping a device, proceed as follows.

1. Open a Windows Command Prompt window.
2. At the command prompt, type, nslookup command and then type DNS of a site , as shown below.

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```
Command Prompt - nslookup
Microsoft Windows [Version 10.0.19044.1526]
(c) Microsoft Corporation. All rights reserved.

C:\Users\LENOVO>nslookup
Default Server: UnKnown
Address: 192.168.43.1

> www.google.com
Server: UnKnown
Address: 192.168.43.1

Non-authoritative answer:
Name: www.google.com
Addresses: 2404:6800:4007:814::2004
          216.58.196.164

>
```

Understanding results:

Observe the above result indicates IP address of www.google.com

Expt.No-29: Identify IPv6 Addresses

- a) Identify the different types of IPv6 addresses.
- b) Examine a host IPv6 network interface and address.
- c) Practice IPv6 address abbreviation.

a) Identify the different types of IPv6 addresses.

Step 1: Review the different types of IPv6 addresses.

- An IPv6 address is 128 bits long. It is most often presented as 32 hexadecimal characters.
- Each hexadecimal character is the equivalent of 4 bits ($4 \times 32 = 128$).
- A non-abbreviated IPv6 host address is shown here:

2001:0DB8:0001:0000:0000:0000:0000:0001

- A hextet is the hexadecimal, IPv6 version of an IPv4 octet. An IPv4 address is 4 octets long, separated by dots. An IPv6 address is 8 hextets long, separated by colons.
- An IPv4 address is 4 octets and is commonly written or displayed in decimal notation.

255.255.255.255

- An IPv6 address is 8 hextets and is commonly written or displayed in hexadecimal notation.
- FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF**
- In an IPv4 address, each individual octet is 8 binary digits (bits). Four octets equals one 32-bit IPv4 address.

11111111 = 255

11111111.11111111.11111111.11111111 = 255.255.255.255

- In an IPv6 address, each individual hextet is 16 bits long. Eight hextets equals one 128-bit IPv6 address.

1111111111111111 = FFFF

1111111111111111.1111111111111111.1111111111111111.1111111111111111.

1111111111111111.1111111111111111.1111111111111111.1111111111111111 =

FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF

- If we read an IPv6 address starting from the left, the first (or far left) hextet identifies the IPv6 address type. For example, if the IPv6 address has all zeros in the far left hextet, then the address is possibly a **loopback address**.

0000:0000:0000:0000:0000:0000:0000:0001 = loopback address

::1 = loopback address abbreviated

- As another example, if the IPv6 address has FE80 in the first hextet, then the address is a **link-local address**.

FE80:0000:0000:0000:C5B7:CB51:3C00:D6CE = link-local address

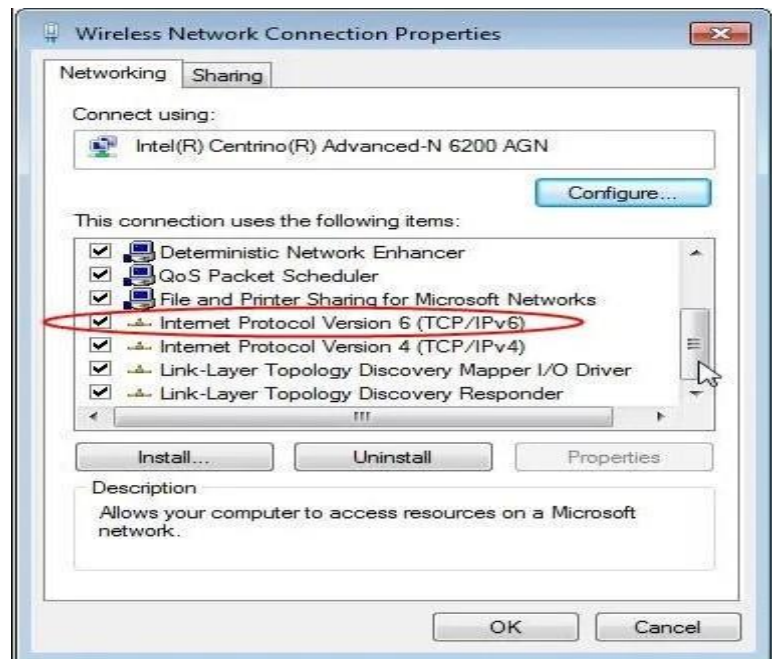
FE80::C5B7:CB51:3C00:D6CE = link-local address abbreviated

b) Examine a host IPv6 network interface and address.

Step 1: Check your PC IPv6 network address settings.

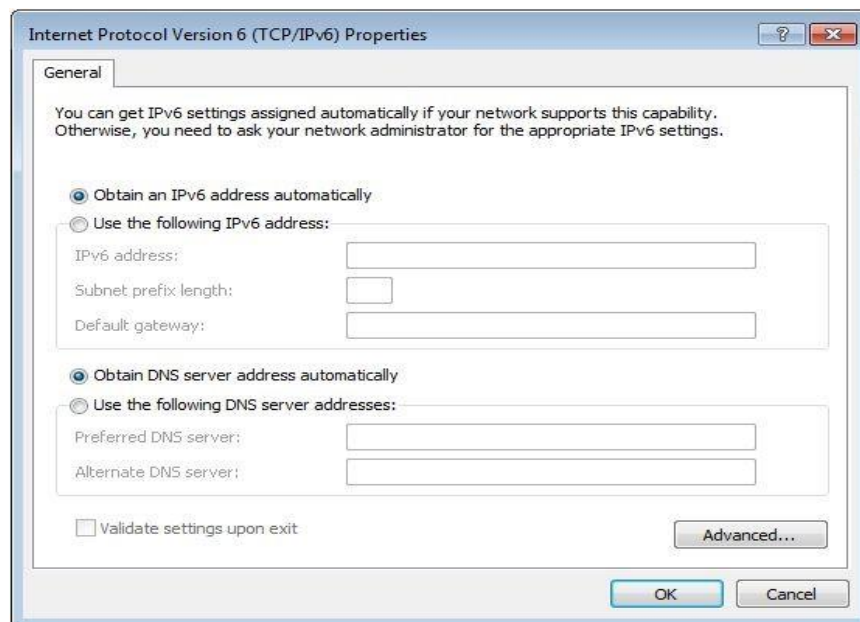
- a. Verify that the IPv6 protocol is installed and active on your PC-A (check your Local Area Connection settings).
- b. Go to **Control Panel** and Click the **Network and Sharing Center** icon.
- c. On the left side of the window, click **Change adapter settings**. There we see icons representing installed network adapters. Double-click on active network interface (it may be a **Local Area Connection** or a **Wireless Network Connection**), and then click **Properties**.

- d. You should now see your Network Connection Properties window. Scroll through the list of items to determine whether **IPv6 is present**, which indicates that it is installed, and if it is also check marked, which indicates that it is active.



- e. Select the item **Internet Protocol Version 6 (TCP/IPv6)** and click **Properties**. You should see the IPv6 settings for your network interface. Your IPv6 properties window is likely set to **Obtain an IPv6 address automatically**. This does not mean that IPv6 relies on the Dynamic Host Configuration Protocol (DHCP). Instead of using DHCP, IPv6 looks to the local router for IPv6 network information and then auto-configures its own IPv6 addresses.
- f. To manually configure IPv6, you must provide the IPv6 address, the subnet prefix length, and the default gateway.

Note: The local router can refer host requests for IPv6 information, especially Domain Name System (DNS) information, to a DHCPv6 server on the network.



- g. After you have verified that IPv6 is installed and active on your PC, you should check your IPv6 address information. To do this, go to **cmd** command prompt window.

- h. Type **ipconfig /all** and press Enter. Your output should look similar to this


```
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix  . : 
Description . . . . . : Realtek RTL8723BE Wireless LAN 802.11n PCI-E NIC
Physical Address. . . . . : 40-9F-38-60-14-33
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2409:4071:4e80:6464:40f6:acdb:53c7:3cb8(Preferred)
Temporary IPv6 Address. . . . . : 2409:4071:4e80:6464:30aa:6341:b615:4302(Preferred)
Link-local IPv6 Address . . . . . : fe80::40f6:acdb:53c7:3cb8%9(Preferred)
IPv4 Address. . . . . : 192.168.43.190(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 27 February 2022 18:17:41
Lease Expires . . . . . : 27 February 2022 20:24:40
Default Gateway . . . . . : fe80::c0e1:49ff:fe23:1499%9
                             192.168.43.1
DHCP Server . . . . . : 192.168.43.1
DHCPv6 IAID . . . . . : 54566712
DHCPv6 Client DUID. . . . . : 00-01-00-01-21-BE-7B-21-80-EE-73-D4-DA-43
DNS Servers . . . . . : 192.168.43.1
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . : 
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 40-9F-38-60-14-32
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

C:\Users\User>
```

c) Practice IPv6 address abbreviation.

Rule 1: In an IPv6 address, a string of four zeros (0s) in a hextet can be abbreviated as a single zero.

2001:0404:0001:1000:0000:0000:0EF0:BC00

2001:0404:0001:1000:0:0:0EF0:BC00 (abbreviated with single zeros)

Rule 2: In an IPv6 address, the leading zeros in each hextet can be omitted, trailing zeros cannot be omitted.

2001:0404:0001:1000:0000:0000:0EF0:BC00

2001:404:1:1000:0:0:EF0:BC00 (abbreviated with leading zeros omitted)

Rule 3: In an IPv6 address, a single continuous string of four or more zeros can be abbreviated as a double colon (::). The double colon abbreviation can only be used one time in an IP address.

2001:0404:0001:1000:0000:0000:0EF0:BC00

2001:404:1:1000::EF0:BC00 (abbreviated with leading zeroes omitted and continuous zeros replaced with a double colon)

The image below illustrates these rules of IPv6 address abbreviation:

```
FF01:0000:0000:0000:0000:0000:0000:1
= FF01:0:0:0:0:0:0:1
= FF01::1
```

```
E3D7:0000:0000:0000:51F4:00C8:C0A8:6420
= E3D7::51F4:C8:C0A8:6420
```

```
3FFE:0501:0008:0000:0260:97FF:FE40:EFAB
= 3FFE:501:8:0:260:97FF:FE40:EFAB
= 3FFE:501:8::260:97FF:FE40:EFAB
```

Expt.No-30 Use simulator to demonstrate Telnet

Objectives

1. Develop an understanding of the basic functions of Packet Tracer.
2. Model a simple network and observe traffic behaviour on the network.
3. Create a simple Ethernet network using 5 hosts as client.

Observe data flow using pings

Background / Preparation

Create a logical network diagram with 5 PCs, connected with straight-through cables.

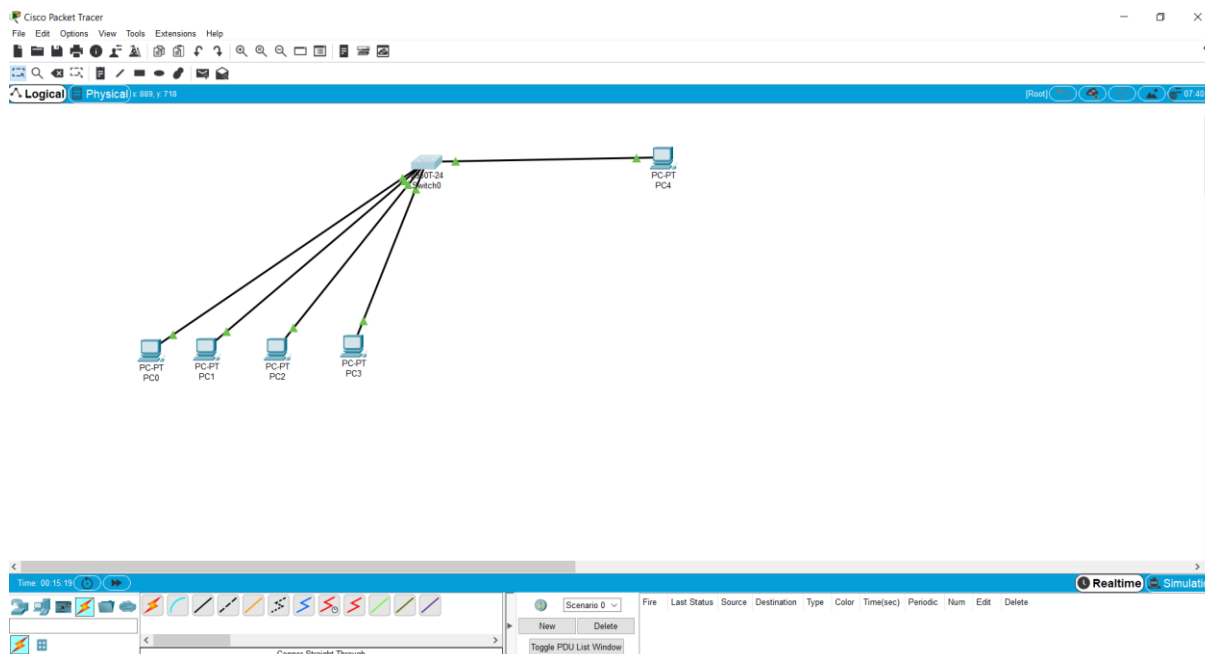
Step 1: Build the network

The bottom left hand corner of the screen displays the icons that represent device categories or groups, such as Routers, Switches, or End Devices. Moving the cursor over the device categories will show options to the right. To select a device, first select the device category and then select the option that is required.

1. Select **End Devices** from the options in the bottom left-hand corner. Drag and drop 5 **Generic PCs** onto your design area.

2. Select **Connections** from the bottom left-hand corner and choose **Copper Straight-Through** connect each PC to the switch.

1. PC0 connects to Fast Ethernet 0/1
2. PC1 connects to Fast Ethernet 0/2
3. PC2 connects to Fast Ethernet 0/3
4. PC3 connects to Fast Ethernet 0/4
5. PC4 connects to Fast Ethernet 0/5



 Switch0

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vlan 1
Switch(config-if)#ip address 192.168.10.100 255.255.255.0
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed
state to up

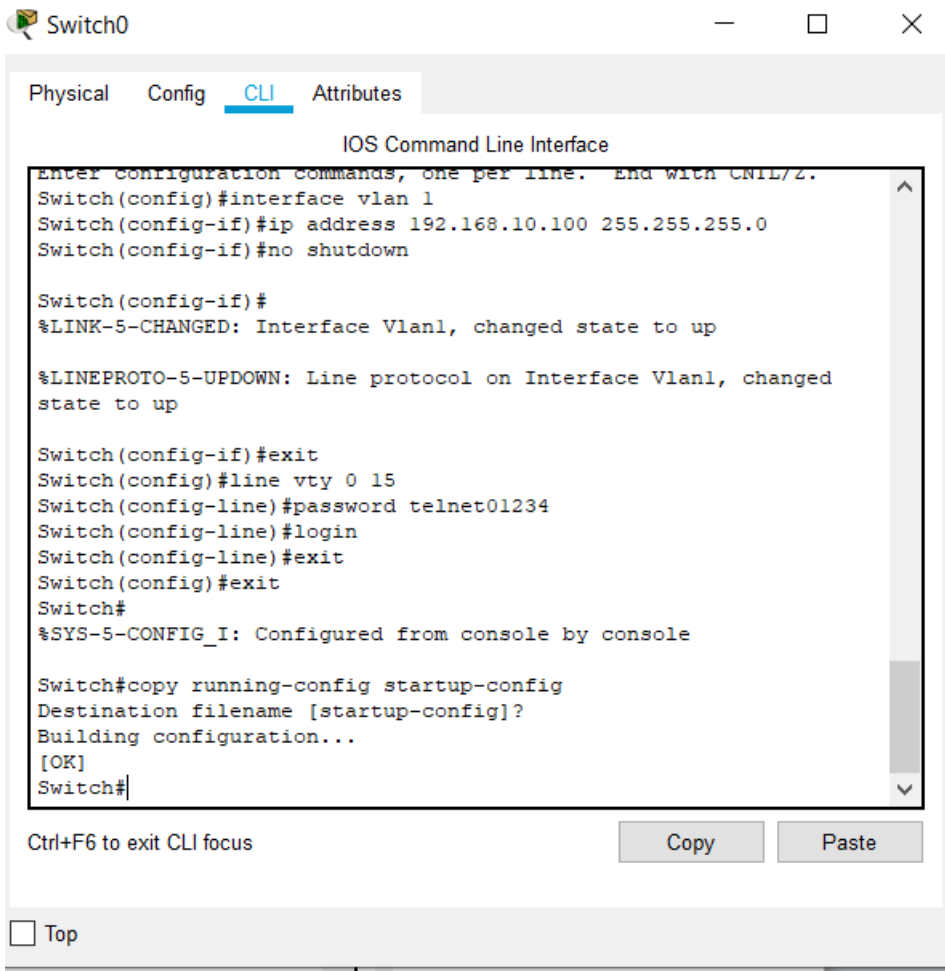
Switch(config-if)#exit
Switch(config)#line vty 0 15
Switch(config-line)#password telnet01234
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#exit
Switch#
```

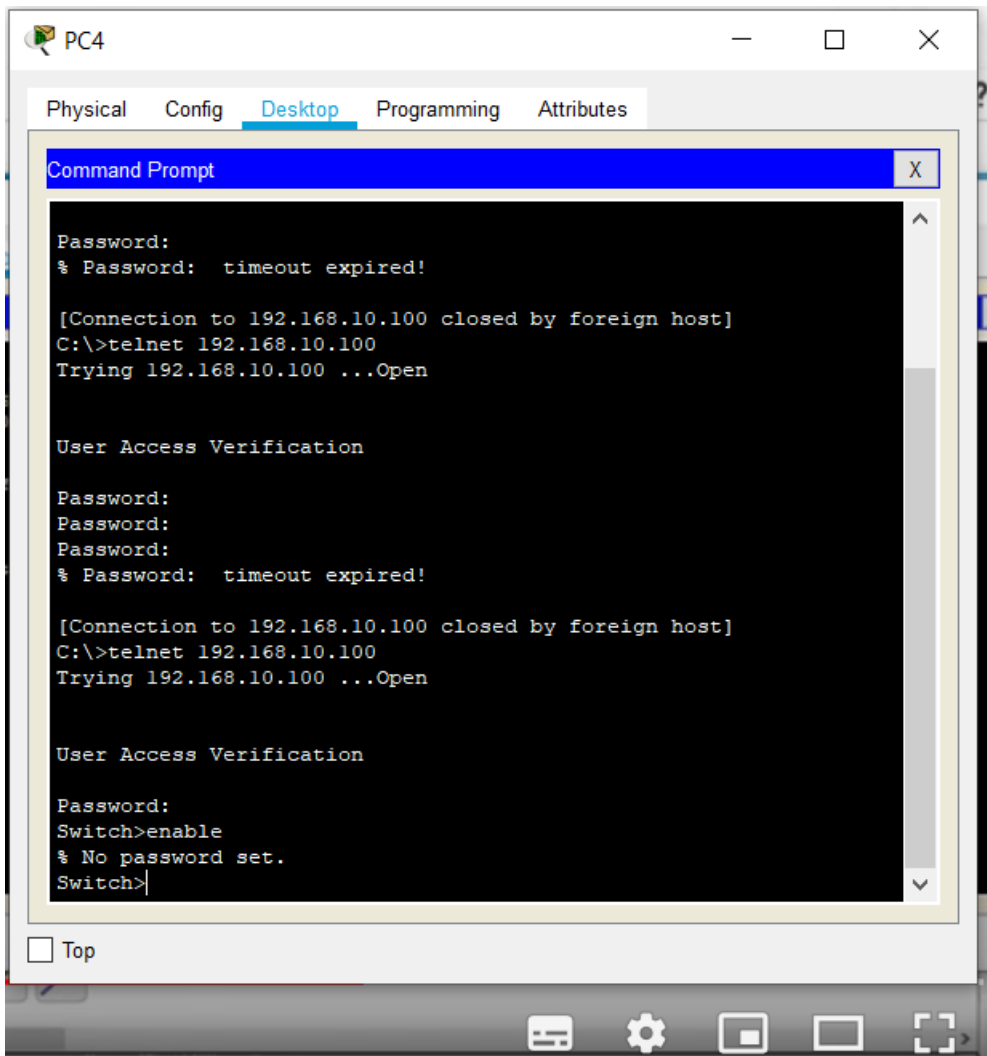
Ctrl+F6 to exit CLI focus

Copy

Paste

☐ Top





The screenshot shows a configuration window for a PC named 'PC4'. The window has four tabs: 'Physical', 'Config', 'Desktop', and 'Attributes'. The 'Desktop' tab is selected. Inside the 'Desktop' tab, there is a sub-tab 'IP Configuration' which is also selected. Below this, the 'Interface' is set to 'FastEthernet0'. The 'IP Configuration' section has two radio buttons: 'DHCP' (unselected) and 'Static' (selected). Below these are four text input fields: 'IPv4 Address' (192.168.10.2), 'Subnet Mask' (255.255.255.0), 'Default Gateway' (0.0.0.0), and 'DNS Server' (0.0.0.0). The 'IPv6 Configuration' section has two radio buttons: 'Automatic' (unselected) and 'Static' (selected). Below these are four text input fields: 'IPv6 Address' (empty), 'Link Local Address' (FE80::201:63FF:FE01:8A36), 'Default Gateway' (empty), and 'DNS Server' (empty). The '802.1X' section has a checkbox 'Use 802.1X Security' (unchecked). The 'Authentication' dropdown is set to 'MD5'. At the bottom left, there is a 'Top' button.

PC4

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.2

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:63FF:FE01:8A36

Default Gateway

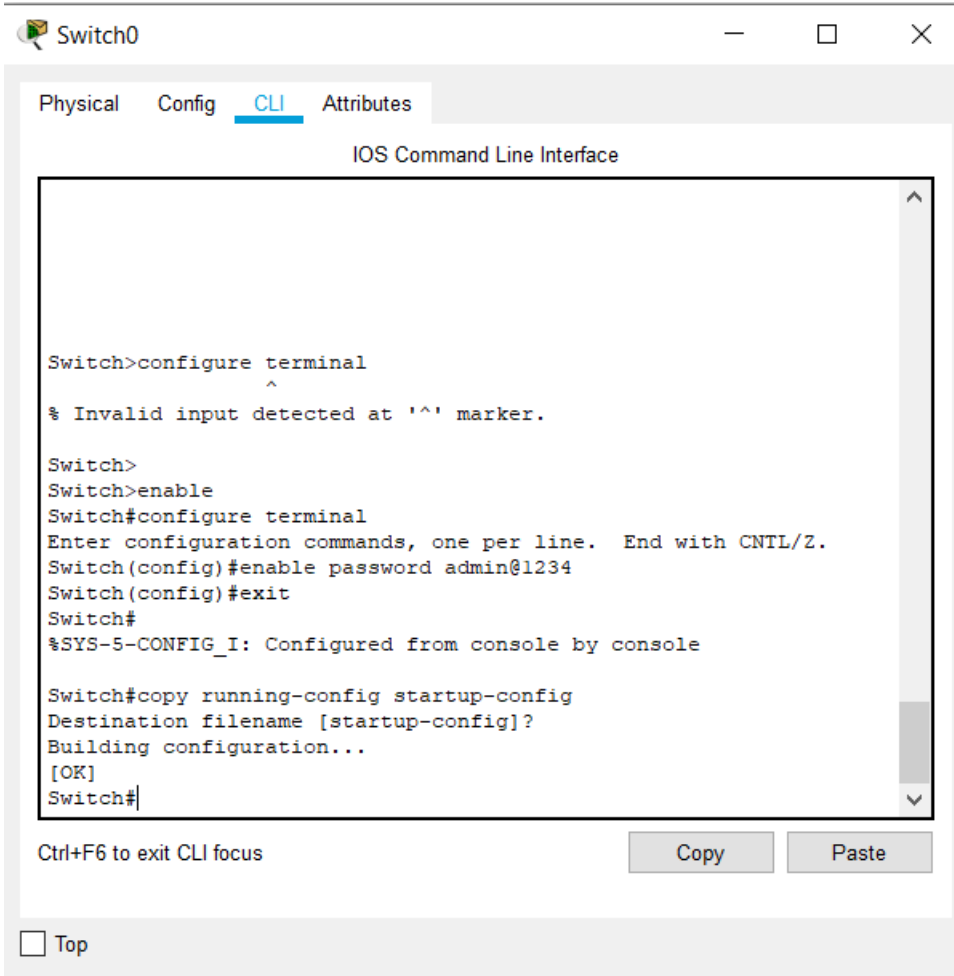
DNS Server

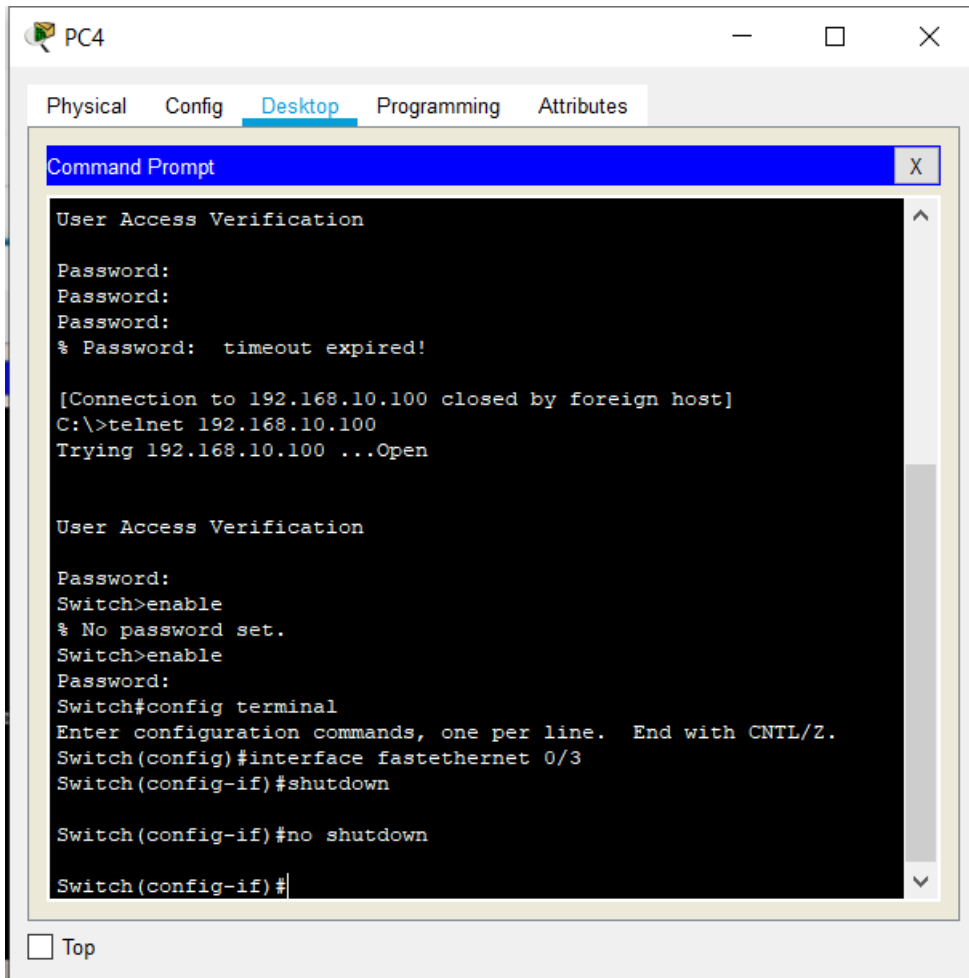
802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top





Expt.No-31: Demonstrate troubleshooting Commands with a scenario- ipconfig, ping, netstat, tracert, nslookup.

i) ipconfig or ipconfig/all

- The “ipconfig” displays the current information about your network such as your device IP and MAC address, and the IP address of your router.
- It can also display information about your DHCP and DNS servers.
- To see the output of ipconfig command, go to command prompt and type it.

```
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix  . : 
IPv6 Address. . . . . : 2401:4900:4bc3:65f8:40f6:acdb:53c7:3cb8
Temporary IPv6 Address. . . . . : 2401:4900:4bc3:65f8:d1da:3f9b:7a6e:360f
Link-local IPv6 Address . . . . . : fe80::40f6:acdb:53c7:3cb8%9
IPv4 Address. . . . . : 192.168.43.234
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::a25:25ff:fe57:e087%9
                             192.168.43.1
```

ii) ping

- The “ping” command ping command allows you to send a signal to another device, and if that device is active, it will send a response back to the sender.
- The “ping” command is a subset of the ICMP (Internet Control Message Protocol), and it uses what is called an “echo request”. So, when you ping a device you send out an echo request, and if the device you pinged is active or online, you get an echo response.
- For example, if your local computer has Internet connectivity issues, you can try to ping your router. If you get no response then you know that the router is what is giving you problems.

```
Cal Command Prompt

C:\Users\Marko>ping 192.168.8.1

Pinging 192.168.8.1 with 32 bytes of data:
Reply from 192.168.8.1: bytes=32 time=1ms TTL=64
Reply from 192.168.8.1: bytes=32 time=16ms TTL=64
Reply from 192.168.8.1: bytes=32 time=20ms TTL=64
Reply from 192.168.8.1: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.8.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 20ms, Average = 9ms

C:\Users\Marko>
```

iii) netstat:

- The network statistics (netstat) command is a networking tool used for troubleshooting and configuration, that can also serve as a monitoring tool for connections over the network.
- Both incoming and outgoing connections, routing tables, port listening, and usage statistics are common uses for this command.

```
C:\> Command Prompt - netstat

Microsoft Windows [Version 10.0.19044.1526]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>netstat

Active Connections

    Proto Local Address          Foreign Address         State
    TCP    192.168.43.234:58157    20.197.71.89:https      ESTABLISHED
    TCP    192.168.43.234:59493    52.244.33.223:https      ESTABLISHED
    TCP    192.168.43.234:60072    13.107.18.254:https      ESTABLISHED
    TCP    192.168.43.234:60507    cloudproxy10041:http     TIME_WAIT
    TCP    192.168.43.234:60544    server-52-84-11-122:https TIME_WAIT
    TCP    192.168.43.234:60552    162.247.243.146:https    TIME_WAIT
    TCP    192.168.43.234:60553    e017:https               TIME_WAIT
    TCP    192.168.43.234:60554    e017:https               TIME_WAIT
    TCP    192.168.43.234:60555    ec2-18-214-172-65:https  TIME_WAIT
    TCP    192.168.43.234:60556    ec2-18-214-172-65:https  TIME_WAIT
```

iv) *tracert*:

- This command lets you see all steps a packet takes to the destination. For example, if we send a packet to www.google.com, it actually goes through a couple of routers to reach the destination.
- The packet will first go to your router, and then it will go to all kinds of different routers before it reaches Google servers. We can also use the term “hops” instead of routers. Let’s run the command and see what kind of results we get.

```
C:\Users\User>tracert 8.8.8.8

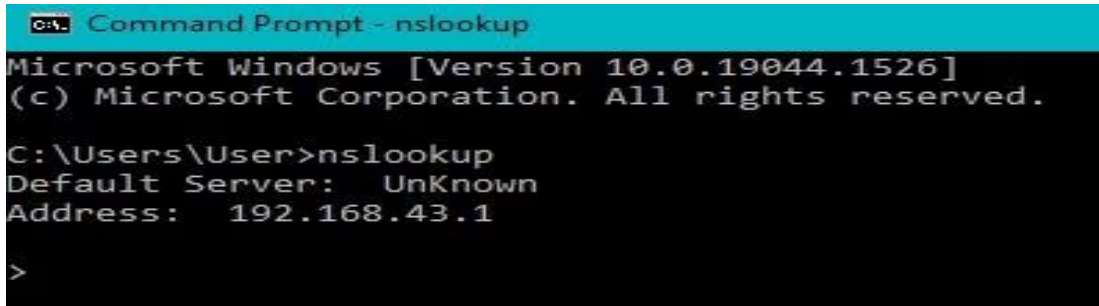
Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  0  5 ms  3 ms  5 ms  192.168.43.1
  1  *      *      *      Request timed out.
  2  77 ms  *      113 ms 10.61.68.65
  3  196 ms 30 ms  39 ms 10.61.66.144
  4  *      *      *      Request timed out.
  5  *      *      *      Request timed out.
  6  *      *      *      Request timed out.
  7  *      *      *      Request timed out.
  8  *      *      *      Request timed out.
  9  *      *      *      Request timed out.
 10  *      *      *      Request timed out.
 11  *      *      *      Request timed out.
 12  *      *      *      Request timed out.
 13  *      590 ms 65 ms dns.google [8.8.8.8]

Trace complete.
```

v) *nslookup*:

- The nslookup command will fetch the DNS records for a given domain name or an IP address.
- Remember the IP addresses and domain names are stored in DNS servers, so the nslookup command lets you query the DNS records to gather information.



```
Command Prompt - nslookup
Microsoft Windows [Version 10.0.19044.1526]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>nslookup
Default Server: UnKnown
Address: 192.168.43.1
>
```